



OPEN A split-step finite element method for the space-fractional Schrödinger equation in two dimensions

Xiaogang Zhu^{1✉}, Haiyang Wan² & Yaping Zhang¹

In this article, we propose a split-step finite element method (FEM) for the two-dimensional nonlinear Schrödinger equation (NLS) with Riesz fractional derivatives in space. The space-fractional NLS is first spatially discretized by finite element scheme and the semi-discrete variational scheme is obtained. We prove that it maintains the mass and energy conservation laws. Then, we establish a fully discrete split-step finite element scheme for the considered problem, which avoids the iteration at each time layer, thereby significantly reducing computational cost. The discrete mass conservation property and error estimate of this split-step finite element scheme is derived. Finally, illustrative tests and the numerical simulation of dynamic of wave solutions are included to confirm its effectiveness and capability.

Keywords Space-fractional Schrödinger equation, Split-step scheme, Finite element scheme, Error estimate

The standard NLS is an important fundamental equation in quantum mechanics, with applications in various fields, such as super-fluids, nonlinear optics, atomic physics, nuclear physics, and solid state physics, which describes the change law of the state of microscopic particles over time by wave function. The one-dimensional NLS has the following form

$$i\hbar \frac{\partial \Psi(x, t)}{\partial t} = -\frac{\hbar^2}{2m} \nabla^2 \Psi(x, t) - \lambda |\Psi(x, t)|^2 \Psi(x, t) + V(x) \Psi(x, t), \quad (1)$$

where $i = \sqrt{-1}$, m is the mass of microscopic particle, $\hbar$ is the Planck's constant, $\lambda |\Psi(x, t)|^2 \Psi(x, t)$ stands for the nonlinear interaction of particles and $V(x)$ is the potential term function. Recently, with the rise of fractional calculus modeling, fractional quantum mechanics have gradually aroused people's interest and the Feynman path integral over the trajectories of Lévy flights leads to the space-fractional NLS^{1,2}, which can characterize more extensive nonrelativistic quantum mechanical behavior in physics. The global well-posedness of fractional NLS was studied by³, the existence and uniqueness of its global solution were proved in^{4,5} and the regularity of ground state was considered by⁶. For some interesting applications in physics, we refer the readers to⁷⁻⁹ for references.

In this article, we consider the Riesz space-fractional NLS in two-dimensions:

$$i \frac{\partial \Psi(x, y, t)}{\partial t} + \gamma \frac{\partial^{2\alpha} \Psi(x, y, t)}{\partial |x|^{2\alpha}} + \gamma \frac{\partial^{2\alpha} \Psi(x, y, t)}{\partial |y|^{2\alpha}} + \lambda |\Psi(x, y, t)|^2 \Psi(x, y, t) = 0, \quad (x, y, t) \in \Omega \times (0, T], \quad (2)$$

subjected to

$$\Psi(x, y, 0) = \Psi_0(x, y), \quad (x, y) \in \mathbb{R}^2, \quad (3)$$

$$\Psi(x, y, t) = 0, \quad (x, y, t) \in \mathbb{R}^2 \setminus \Omega \times (0, T], \quad (4)$$

where $1/2 < \alpha \leq 1$, $\lambda, \gamma \in \mathbb{R}^+$, $\Omega = (a, b) \times (c, d)$ and $\Psi_0(x, y)$ is a known function vanishing outside Ω . If $\alpha = 1$, it reduces to the classical NLS. The space-fractional derivatives are defined in Riesz sense by

¹School of Science, Shaoyang University, Shaoyang 422000, Hunan, People's Republic of China. ²School of Future Technology, South China University of Technology, Guangzhou 510641, Guangdong, People's Republic of China. ✉email: zhuxg590@yeah.net

$$\begin{aligned}\frac{\partial^{2\alpha}\Psi(x,y,t)}{\partial|x|^{2\alpha}} &= \frac{-1}{2\cos(\alpha\pi)}\left\{ {}^L D_x^{2\alpha}\Psi(x,y,t) + {}^R D_b^{2\alpha}\Psi(x,y,t) \right\}, \\ {}^L D_x^{2\alpha}\Psi(x,y,t) &= \frac{1}{\Gamma(2-2\alpha)}\frac{\partial^2}{\partial x^2}\int_a^x(x-\xi)^{1-2\alpha}\Psi(\xi,y,t)d\xi, \\ {}^R D_b^{2\alpha}\Psi(x,y,t) &= \frac{1}{\Gamma(2-2\alpha)}\frac{\partial^2}{\partial x^2}\int_x^b(\xi-x)^{1-2\alpha}\Psi(\xi,y,t)d\xi,\end{aligned}$$

and $\frac{\partial^{2\alpha}\Psi(x,y,t)}{\partial|y|^{2\alpha}}$ can be defined in the same fashion.

Meanwhile, the Eqs. (2)–(4) have the mass and energy conservative laws. Define the below functionals

$$\begin{aligned}M(t) &= \int_{\Omega} |\Psi(x,y,t)|^2 d\Omega, \\ E(t) &= \int_{\Omega} \left| \frac{\partial^\alpha \Psi(x,y,t)}{\partial|x|^\alpha} \right|^2 + \left| \frac{\partial^\alpha \Psi(x,y,t)}{\partial|y|^\alpha} \right|^2 d\Omega - \frac{\lambda}{2} \int_{\Omega} |\Psi(x,y,t)|^4 d\Omega,\end{aligned}$$

then there exist $M(t) = M(0)$, $E(t) = E(0)$, $0 < t < T^{10}$.

The past decade has witnessed the rapid evolution of numerical algorithms for the space-fractional NLS, although it is quite complex to solve the space-fractional partial differential equations (PDEs). Aside from a few analytic and semi-analytical techniques^{11,12}, Wang et al. developed an efficient finite difference method (FDM) for the Riesz space-fractional coupled NLS (CNLS)¹³. Ran and Zhang proposed an implicit FDM for the space-fractional strongly CNLS¹⁴, which preserves the discrete conservative laws. A Crank-Nicolson FDM was considered for the Riesz space-fractional NLS¹⁵, which preserves the discrete energy. In^{16,17}, Li et al. developed a fast linearized FEM for the space-fractional NLS and strongly CNLS, respectively. In^{18,19}, we proposed an implicit and a linearized FEM for the time-space-fractional NLS and the space-fractional CNLS, respectively. As compared to the one-dimensional space-fractional problems, it is more difficulty to design numerical methods to solve the multi-dimensional fractional problems due to the complex vector structure of fractional Laplacian operator. The numerical methods for the multi-dimensional fractional PDEs face great challenge in the aspect of algorithm design and computational efficiency because of the non-locality of fractional derivative. In spite of this, there still exist a few numerical works carried out for the two-dimensional space-fractional NLS. Zhao et al. proposed a compact alternating direction implicit (ADI) difference scheme for the two-dimensional space-fractional NLS²⁰. Wang and Huang developed a split-step ADI FDM¹⁰, which conserves the discrete mass. Fan and Qi introduced an efficient FEM for the two-dimensional time-space-fractional NLS²¹. Zhang et al. constructed a conservative FDM²², which utilizes the compact implicit integration factor method to discretize the resulting nonlinear ordinary differential system. Wang et al. proposed a split-step spectral Galerkin method for the two-dimensional space-fractional NLS and proved that it preserves the discrete mass²³. Wang et al. derived a Lie-Trotter operator splitting spectral method for the linear semiclassical fractional Schrödinger equation²⁴. Abdolabadi et al. presented a split-step Fourier pseudo-spectral method for the space-fractional CNLS²⁵.

Recently, the numerical algorithms for the nonlinear multi-dimensional space-fractional problems are of current attention. For these types of equation, how to discretize the nonlinear terms is the key point to design an efficient algorithm. Among various techniques, the split-step method can avoid internal iteration loop at each time level by dividing the original problems into nonlinear term and linear derivative term by time splitting^{26,27}, it can accelerate the computational speed in practice, hence it is an effective approach to solve the complex nonlinear problems. Meanwhile, in multi-physical fields analysis, the FEM is recognized as a well-known numerical technique to obtain the solutions of complex fractional boundary value problems. Since the FDM always adopts difference operators to approximating the fractional derivatives, it has a relatively high requirement for the regularity of the solutions. For example, the compact difference operator constructed in²⁰ requires the unknown function $f(x) \in C^7(\mathbb{R})$ and it is usually difficult to meet this criteria for the fractional PDEs in practice, so are the spectral methods. Although the FDM would be a little cheaper than FEM, the FEM can achieve high-order convergence with much lower smoothness requirement for the solution. However, since it is based on weak form and the numerical algorithm of space-fractional models is highly complex, few research work was reported on the FEM for multi-dimensional space-fractional NLS. This work is devoted to constructing a conservative FEM to solve the space-fractional NLS in two-dimensions. The original problem is splitted into linear and nonlinear subproblems via a split-step technique and the subproblems are solved in a given sequential order. We establish a conservative FEM for the linear subproblem with the time derivative discretized by Crank-Nicolson scheme and detailed analysis on its convergence would be done. The proposed split-step FEM can effectively reduce computational cost and algorithm complexity because no iteration is required. We prove that this split-step scheme possesses the mass conservative property and carry out an analysis for its convergence as well.

The remainder is as follows. In Section 2, we introduce some preliminaries on the fractional derivative and fractional Sobolev spaces. In Section 3–4, the variational formula and split-step FEM are reported and in Section 5, we discuss its mass conservation property and convergence. In Section 6, illustrative examples and the numerical simulation of wave dynamics are performed to check its excellent efficiency. Lastly, we end the study with some conclusive remarks.

Fractional derivative spaces

To begin with, we require some results about fractional derivatives, which have been studied by^{28,29}. Letting

$$\|u\|_{0,\Omega} = (u, u)^{1/2}, \quad (u, v) = \int_c^d \int_a^b u(x, y) v(x, y) dx dy,$$

we introduce the following fractional derivative spaces and lemmas.

Definition 2.1 (Left Fractional Derivative Space) For $\mu > 0$, define the left semi-norm

$$|u|_{J_L^\mu(\Omega)} = (|{}_a^L D_x^\mu u|_{0,\Omega}^2 + |{}_c^L D_y^\mu u|_{0,\Omega}^2)^{1/2},$$

and left norm

$$\|u\|_{J_L^\mu(\Omega)} = (\|u\|_{0,\Omega}^2 + |u|_{J_L^\mu(\Omega)}^2)^{1/2},$$

and let $J_{L,0}^\mu(\Omega)$ be the closure of $C_0^\infty(\Omega)$ with respect to $\|\cdot\|_{J_L^\mu(\Omega)}$.

Definition 2.2 (Right Fractional Derivative Space) For $\mu > 0$, define the right semi-norm

$$|u|_{J_R^\mu(\Omega)} = (|{}_x^R D_b^\mu u|_{0,\Omega}^2 + |{}_y^R D_d^\mu u|_{0,\Omega}^2)^{1/2},$$

and right norm

$$\|u\|_{J_R^\mu(\Omega)} = (\|u\|_{0,\Omega}^2 + |u|_{J_R^\mu(\Omega)}^2)^{1/2},$$

and let $J_{R,0}^\mu(\Omega)$ be the closure of $C_0^\infty(\Omega)$ with respect to $\|\cdot\|_{J_R^\mu(\Omega)}$.

Definition 2.3 (Fractional Sobolev Space) For $\mu > 0$, define the semi-norm

$$|u|_{\mu,\Omega} = \|\xi^\mu \mathcal{F}[\tilde{u}]\|_{0,\Omega},$$

and the norm

$$\|u\|_{\mu,\Omega} = (\|u\|_{0,\Omega}^2 + |u|_{\mu,\Omega}^2)^{1/2},$$

and denote the closure of $C_0^\infty(\Omega)$ with respect to $\|\cdot\|_{\mu,\Omega}$ by $H_0^\mu(\Omega)$, where $\mathcal{F}[\tilde{u}]$ is the Fourier transform of the extension of u by zero outside of Ω .

Lemma 2.1^{29,30} Let $\mu \neq n - 1/2$, $n \in \mathbb{Z}^+$, $J_{L,0}^\mu(\Omega)$, $J_{R,0}^\mu(\Omega)$ and $H_0^\mu(\Omega)$ are equivalent with the equivalent semi-norms and norms, i.e.,

$$C_1 \|u\|_{\mu,\Omega} \leq \max \{ \|u\|_{J_L^\mu(\Omega)}, \|u\|_{J_R^\mu(\Omega)} \} \leq C_2 \|u\|_{\mu,\Omega},$$

where C_1, C_2 are two positive constants independent of u .

Lemma 2.2²⁸ For $0 < \mu < 1$ and $u, v \in J_{L,0}^{2\mu}(\Omega)$ or $J_{R,0}^{2\mu}(\Omega)$, one has

$$\begin{aligned} ({}_a^L D_x^{2\mu} u, v) &= ({}_a^L D_x^\mu u, {}_x^R D_b^\mu v), \\ ({}_x^R D_b^{2\mu} u, v) &= ({}_x^R D_b^\mu u, {}_a^L D_x^\mu v), \end{aligned}$$

and the similar results can be derived for the fractional derivatives in y -direction.

Lemma 2.3³⁰ For $\mu > 0$, one has

$$\begin{aligned} ({}_a^L D_x^\mu u, {}_x^R D_b^\mu u) + ({}_x^R D_b^\mu u, {}_a^L D_x^\mu u) &= 2 \cos(\mu\pi) \|{}_a^L D_x^\mu \tilde{u}\|_{0,\mathbb{R}^2}^2, \\ ({}_c^L D_y^\mu u, {}_y^R D_d^\mu u) + ({}_y^R D_d^\mu u, {}_c^L D_y^\mu u) &= 2 \cos(\mu\pi) \|{}_c^L D_y^\mu \tilde{u}\|_{0,\mathbb{R}^2}^2. \end{aligned}$$

In a similar way to classical theory, we have the following Poincaré-Friedrichs inequalities.

Lemma 2.4 (Fractional Poincaré-Friedrichs^{28,29}) *For $0 < \gamma < \mu$, one has*

$$\|u\|_{0,\Omega} \leq C|u|_{J_L^\mu(\Omega)}, \quad |u|_{J_L^\gamma(\Omega)} \leq C|u|_{J_L^\mu(\Omega)},$$

and for $0 < \gamma < \mu$, $\mu \neq n - 1/2$, $n \in \mathbb{Z}^+$, one has

$$\|u\|_{0,\Omega} \leq C|u|_{\mu,\Omega}, \quad |u|_{\gamma,\Omega} \leq C|u|_{\mu,\Omega},$$

with a positive constant C independent of u . The similar results can be derived for the right fractional semi-norm and norm.

Semi-discrete variational formula

Assume that $\mathcal{M}_h$ is a uniform partition of Ω with the triangles $\{\Omega_j\}_{j=1}^M$ and $h = \max_{1 \leq j \leq M} \{\text{diam } \Omega_j\}$. Define $\mathcal{V}_h = \{u : u|_{\Omega_j} \in P_{r-1}, \forall \Omega_j \in \mathcal{M}_h\}$ and the finite element subspace $\mathcal{V}_h \subset C(\bar{\Omega}) \cap H_0^\alpha(\bar{\Omega})$, where P_{r-1} denotes the polynomials of degree no more than $r - 1$ on Ω_j . Decomposing the wave function in into its real and imaginary parts by $\Psi(x, y, t) = u(x, y, t) + iv(x, y, t)$. Inserting it into Eqs. (2)–(4), the original problem can be recast as a coupled system

$$\frac{\partial u(x, y, t)}{\partial t} + \gamma \frac{\partial^{2\alpha} v(x, y, t)}{\partial |x|^{2\alpha}} + \gamma \frac{\partial^{2\alpha} v(x, y, t)}{\partial |y|^{2\alpha}} + \lambda(u^2(x, y, t) + v^2(x, y, t))v(x, y, t) = 0, \quad (5)$$

$$\frac{\partial v(x, y, t)}{\partial t} - \gamma \frac{\partial^\alpha u(x, y, t)}{\partial |x|^\alpha} - \gamma \frac{\partial^{2\alpha} u(x, y, t)}{\partial |y|^{2\alpha}} - \lambda(u^2(x, y, t) + v^2(x, y, t))u(x, y, t) = 0, \quad (6)$$

subjected to

$$u(x, y, 0) = \text{Re}\Psi_0(x, y), \quad (7)$$

$$v(x, y, 0) = \text{Im}\Psi_0(x, y), \quad (x, y) \in \mathbb{R}^2, \quad (8)$$

$$u(x, y, t) = 0, \quad v(x, y, t) = 0, \quad (x, y, t) \in \mathbb{R}^2 \setminus \Omega \times (0, T], \quad (9)$$

where “Re”, “Im” mean keeping the real and imaginary parts, respectively.

Using Lemma 2.2, we pose the semi-discrete variational formula to seek $u_h, v_h \in \mathcal{V}_h$ for each t such that

$$\left(\frac{\partial u_h(t)}{\partial t}, \chi_1 \right) - \gamma \Lambda_h(v_h(t), \chi_1) + \lambda(|u_h(t)|^2 + |v_h(t)|^2)v_h(t), \chi_1 = 0, \quad (10)$$

$$\left(\frac{\partial v_h(t)}{\partial t}, \chi_2 \right) + \gamma \Lambda_h(u_h(t), \chi_2) - \lambda(|u_h(t)|^2 + |v_h(t)|^2)u_h(t), \chi_2 = 0, \quad (11)$$

$$u_h(0) = u_0(x, y), \quad v_h(0) = v_0(x, y), \quad (12)$$

for $\chi_1, \chi_2 \in \mathcal{V}_h$, with the bilinear form

$$\Lambda_h(u, v) = \frac{1}{2 \cos(\alpha\pi)} \left[({}_a^L D_x^\alpha u, {}_x^R D_b^\alpha v) + ({}_x^R D_b^\alpha u, {}_a^L D_x^\alpha v) \right] + \frac{1}{2 \cos(\alpha\pi)} \left[({}_c^L D_y^\alpha u, {}_y^R D_d^\alpha v) + ({}_y^R D_d^\alpha u, {}_c^L D_y^\alpha v) \right],$$

where $u_0(x, y), v_0(x, y)$ are some approximations of $\text{Re}\Psi_0(x, y)$ and $\text{Im}\Psi_0(x, y)$, respectively.

Theorem 3.1³¹ *The bilinear form $\Lambda_h(\cdot, \cdot)$ satisfies*

$$\begin{aligned} \Lambda_h(u, v) &\leq C \|u\|_{\text{eng}} \|v\|_{\text{eng}}, \\ \Lambda_h(u, u) &\geq C \|u\|_{\text{eng}}^2, \end{aligned}$$

with the associated energy norm as

$$\begin{aligned} \|u\|_{eng} &= (\|u\|_{0,\Omega}^2 + |u|_{eng}^2)^{1/2}, \\ |u|_{eng} &= \left(C_\alpha |(L_a D_x^\alpha u, {}^R D_b^\alpha u) + ({}^R D_b^\alpha u, L_a D_x^\alpha u)| + C_\alpha |(L_c D_y^\alpha u, {}^R D_d^\alpha u) + ({}^R D_d^\alpha u, L_c D_y^\alpha u)| \right)^{1/2}, \end{aligned}$$

where $C_\alpha = \frac{1}{2\cos(\alpha\pi)}$ and C is a positive constant independent of u, v .

Theorem 3.2 The variational formula (10)–(12) preserves the mass and energy conservation laws, i.e.,

$$\frac{dM_h(t)}{dt} = 0, \quad \frac{dE_h(t)}{dt} = 0,$$

where

$$\begin{aligned} M_h(t) &= \|u_h(t)\|_{0,\Omega}^2 + \|v_h(t)\|_{0,\Omega}^2, \\ E_h(t) &= \gamma |u_h(t)|_{eng}^2 + \gamma |v_h(t)|_{eng}^2 - \frac{\lambda}{2} \|u_h^2(t) + v_h^2(t)\|_{0,\Omega}^2. \end{aligned}$$

Proof Letting $\chi_1 = u_h(t)$ in Eq. (10) and $\chi_2 = v_h(t)$ in Eq. (11), adding both two equations together and using the symmetry property of $\Lambda_h(\cdot, \cdot)$, we have

$$\left(\frac{\partial u_h(t)}{\partial t}, u_h(t) \right) + \left(\frac{\partial v_h(t)}{\partial t}, v_h(t) \right) = \frac{1}{2} \frac{d}{dt} \|u_h(t)\|_{0,\Omega}^2 + \frac{1}{2} \frac{d}{dt} \|v_h(t)\|_{0,\Omega}^2 = 0,$$

which implies $\frac{dM_h(t)}{dt} = 0$.

Furthermore, taking $\chi_1 = \frac{\partial v_h(t)}{\partial t}$, $\chi_2 = \frac{\partial u_h(t)}{\partial t}$ in Eqs. (10) and (11), we have

$$\left(\frac{\partial u_h(t)}{\partial t}, \frac{\partial v_h(t)}{\partial t} \right) - \gamma \Lambda_h \left(v_h(t), \frac{\partial v_h(t)}{\partial t} \right) + \lambda \left((u_h^2(t) + v_h^2(t)) v_h(t), \frac{\partial v_h(t)}{\partial t} \right) = 0, \quad (13)$$

$$\left(\frac{\partial v_h(t)}{\partial t}, \frac{\partial u_h(t)}{\partial t} \right) + \gamma \Lambda_h \left(u_h(t), \frac{\partial u_h(t)}{\partial t} \right) - \lambda \left((u_h^2(t) + v_h^2(t)) u_h(t), \frac{\partial u_h(t)}{\partial t} \right) = 0, \quad (14)$$

and the difference between Eqs. (13) and (14) gives

$$\begin{aligned} \gamma \Lambda_h \left(u_h(t), \frac{\partial u_h(t)}{\partial t} \right) + \gamma \Lambda_h \left(v_h(t), \frac{\partial v_h(t)}{\partial t} \right) \\ = \lambda \left((u_h^2(t) + v_h^2(t)) u_h(t), \frac{\partial u_h(t)}{\partial t} \right) + \lambda \left((u_h^2(t) + v_h^2(t)) v_h(t), \frac{\partial v_h(t)}{\partial t} \right). \end{aligned} \quad (15)$$

On the other hand, from the following identities

$$\begin{aligned} \frac{d(L_a D_x^\alpha u_h(t), {}^R D_b^\alpha u_h(t))}{dt} &= \left(\frac{\partial L_a D_x^\alpha u_h(t)}{\partial t}, {}^R D_b^\alpha u_h(t) \right) + \left(L_a D_x^\alpha u_h(t), \frac{\partial {}^R D_b^\alpha u_h(t)}{\partial t} \right), \\ \frac{d(L_c D_y^\alpha u_h(t), {}^R D_d^\alpha u_h(t))}{dt} &= \left(\frac{\partial L_c D_y^\alpha u_h(t)}{\partial t}, {}^R D_d^\alpha u_h(t) \right) + \left(L_c D_y^\alpha u_h(t), \frac{\partial {}^R D_d^\alpha u_h(t)}{\partial t} \right), \end{aligned}$$

it follows that

$$\frac{d}{dt} \Lambda_h(u_h(t), u_h(t)) = 2\Lambda_h \left(u_h(t), \frac{\partial u_h(t)}{\partial t} \right) = \frac{d}{dt} |u_h(t)|_{eng}^2.$$

In view of the definition of energy norm, we obtain

$$\gamma \Lambda_h \left(u_h(t), \frac{\partial u_h(t)}{\partial t} \right) + \gamma \Lambda_h \left(v_h(t), \frac{\partial v_h(t)}{\partial t} \right) = \frac{\gamma}{2} \frac{d}{dt} |u_h(t)|_{eng}^2 + \frac{\gamma}{2} \frac{d}{dt} |v_h(t)|_{eng}^2. \quad (16)$$

For the right-side term of Eq. (15), we have

$$\begin{aligned} & \lambda \left((u_h^2(t) + v_h^2(t))u_h(t), \frac{\partial u_h(t)}{\partial t} \right) + \lambda \left((u_h^2(t) + v_h^2(t))v_h(t), \frac{\partial v_h(t)}{\partial t} \right) \\ &= \lambda \left(u_h^3(t) + v_h^2(t)u_h(t), \frac{\partial u_h(t)}{\partial t} \right) + \lambda \left(u_h^2(t)v_h(t) + v_h^3(t), \frac{\partial v_h(t)}{\partial t} \right) \\ &= \frac{\lambda}{2} \left(u_h^2(t) + v_h^2(t), 2u_h(t) \frac{\partial u_h(t)}{\partial t} + 2v_h(t) \frac{\partial v_h(t)}{\partial t} \right) \\ &= \frac{\lambda}{2} \frac{d}{dt} (u_h^2(t) + v_h^2(t)) \\ &= \frac{\lambda}{2} \frac{d}{dt} \|u_h^2(t) + v_h^2(t)\|_{0,\Omega}^2. \end{aligned} \quad (17)$$

Substituting Eq. (16) and Eq. (17) into Eq. (15), we obtain

$$\gamma \frac{d}{dt} |u_h(t)|_{eng}^2 + \gamma \frac{d}{dt} |v_h(t)|_{eng}^2 = \frac{\lambda}{2} \frac{d}{dt} \|u_h^2(t) + v_h^2(t)\|_{0,\Omega}^2,$$

and this finally implies $\frac{dE_h(t)}{dt} = 0$. This completes the proof. $\square$

Derivation of split-step FEM for space-fractional NLS

This section devotes to derive a fully discrete split-step finite element scheme for Eqs. (2)–(4). Define a partition on $[0, T]$ with equally spaced grid points $t_k = k\Delta t$, $\Delta t = T/N$, $k = 0, 1, \dots, N$ and $N \in \mathbb{Z}^+$, and for simplicity, we employ

$$\bar{\partial}_t f^{n+1} = \frac{f^{n+1} - f^n}{\Delta t}, \quad \bar{f}^{n+\frac{1}{2}} = \frac{f^{n+1} + f^n}{2}, \quad f^n = f(t_n).$$

If $\Psi(x, y, t_n)$ is known, the wave function at t_{n+1} can be obtained by solving the considered equation in two steps via splitting it into the linear and nonlinear subparts based on the second-order Strang split-step technique³². If the NLS is given by $i\Psi_t = L\Psi + N\Psi$ with $L\Psi$, $N\Psi$ being the linear and nonlinear parts, by separation of variables, we can get

$$\Psi(t_n + \Delta t) = e^{-i\Delta t(L+N)}\Psi(t_n) \approx J_t[\Psi(t_n)] = \mathcal{S}^N \left(\frac{\Delta t}{2} \right) \mathcal{S}^L(\Delta t) \mathcal{S}^N \left(\frac{\Delta t}{2} \right) \Psi(t_n), \quad (18)$$

where $\mathcal{S}^L$, $\mathcal{S}^N$ are the exact solution operators associated with the fractional and nonlinear equations:

$$\begin{aligned} & i \frac{\partial \Psi(x, y, t)}{\partial t} + \gamma \frac{\partial^{2\alpha} \Psi(x, y, t)}{\partial |x|^{2\alpha}} + \gamma \frac{\partial^{2\alpha} \Psi(x, y, t)}{\partial |y|^{2\alpha}} = 0, \\ & i \frac{\partial \Psi(x, y, t)}{\partial t} + \lambda |\Psi(x, y, t)|^2 \Psi(x, y, t) = 0, \end{aligned}$$

which further leads to

$$\frac{\partial u(x, y, t)}{\partial t} + \lambda (u^2(x, y, t) + v^2(x, y, t))v(x, y, t) = 0, \quad (19)$$

$$\frac{\partial v(x, y, t)}{\partial t} - \lambda (u^2(x, y, t) + v^2(x, y, t))u(x, y, t) = 0, \quad (20)$$

and the linear subproblem

$$\frac{\partial u(x, y, t)}{\partial t} + \gamma \frac{\partial^{2\alpha} v(x, y, t)}{\partial |x|^{2\alpha}} + \gamma \frac{\partial^{2\alpha} v(x, y, t)}{\partial |y|^{2\alpha}} = 0, \quad (21)$$

$$\frac{\partial v(x, y, t)}{\partial t} - \gamma \frac{\partial^{2\alpha} u(x, y, t)}{\partial |x|^{2\alpha}} - \gamma \frac{\partial^{2\alpha} u(x, y, t)}{\partial |y|^{2\alpha}} = 0. \quad (22)$$

Consequently, we can achieve the approximate solution of the coupled system (5) and (6) by solving the linear subproblem (21) and (22), followed by the nonlinear subproblem (19) and (20). At first, for the nonlinear part, we can integrate them exactly in physical space. Multiplying both sides of Eq. (19) by $u(x, y, t)$ and Eq. (20) by $v(x, y, t)$, and adding them together lead to $\frac{du^2(x, y, t)}{dt} + \frac{dv^2(x, y, t)}{dt} = 0$, which implies

$$u^2(x, y, t) + v^2(x, y, t) = u^2(x, y, t_n) + v^2(x, y, t_n), \quad t \in (t_n, t_{n+1}]. \quad (23)$$

Replacing $u^2(x, y, t) + v^2(x, y, t)$ by $u^2(x, y, t_n) + v^2(x, y, t_n)$ admits the coupled system:

$$\frac{du(x, y, t)}{dt} = -\lambda |\Psi(t_n)|^2 v(x, y, t), \quad (24)$$

$$\frac{dv(x, y, t)}{dt} = \lambda |\Psi(t_n)|^2 u(x, y, t), \quad (25)$$

with $|\Psi(t_n)|^2 = u^2(x, y, t_n) + v^2(x, y, t_n)$, which are linear ordinary differential equations (ODEs).

In view of Eq. (24), there exist $v(x, y, t) = -\frac{1}{\lambda |\Psi(t_n)|^2} \frac{du(x, y, t)}{dt}$. Substituting it into Eq. (25), we have the second-order homogeneous ODE with constant coefficient

$$\frac{d^2 u(x, y, t)}{dt^2} + \lambda^2 |\Psi(t_n)|^4 u(x, y, t) = 0. \quad (26)$$

Since its characteristic equation of ODE (26) has two different complex roots $r_{1,2} = \pm i\lambda |\Psi(t_n)|^2$, we obtain its general solution, given by

$$u(x, y, t) = \bar{C}_1 \cos [\lambda |\Psi(t_n)|^2 (t - t_n)] + \bar{C}_2 \sin [\lambda |\Psi(t_n)|^2 (t - t_n)]. \quad (27)$$

with the two general constants $\bar{C}_1$ and $\bar{C}_2$.

Letting $t = t_n$ in (27) gives $\bar{C}_1 = u(x, y, t_n)$ and differentiating (27) with respect to t gives

$$\frac{du(x, y, t)}{dt} = -\lambda \bar{C}_1 |\Psi(t_n)|^2 \sin [\lambda |\Psi(t_n)|^2 (t - t_n)] + \lambda \bar{C}_2 |\Psi(t_n)|^2 \cos [\lambda |\Psi(t_n)|^2 (t - t_n)]. \quad (28)$$

Take $t = t_n$ in (28) and the comparison between (28) with (24) implies $\bar{C}_2 = -v(x, y, t_n)$. Substituting $\bar{C}_1, \bar{C}_2$ into (27), we have the exact solution of ODEs (24)–(25), i.e.,

$$u(x, y, t) = \cos [\lambda |\Psi(t_n)|^2 (t - t_n)] u(x, y, t_n) - \sin [\lambda |\Psi(t_n)|^2 (t - t_n)] v(x, y, t_n), \quad (29)$$

and repeating a similar derivation as above, we obtain

$$v(x, y, t) = \sin [\lambda |\Psi(t_n)|^2 (t - t_n)] u(x, y, t_n) + \cos [\lambda |\Psi(t_n)|^2 (t - t_n)] v(x, y, t_n). \quad (30)$$

As for the linear subproblem, we construct a fully discrete FEM, which adopts the Crank–Nicolson scheme in time. Using Lemma 2.2, the scheme reads: seek $u_h^n, v_h^n \in \mathcal{V}_h$ to satisfy

$$(\bar{\partial}_t u_h^{n+1}, \chi_1) - \gamma \Lambda_h(\bar{v}_h^{n+\frac{1}{2}}, \chi_1) = 0, \quad (31)$$

$$(\bar{\partial}_t v_h^{n+1}, \chi_2) + \gamma \Lambda_h(\bar{u}_h^{n+\frac{1}{2}}, \chi_2) = 0, \quad \text{for } \chi_1, \chi_2 \in \mathcal{V}_h, \quad (32)$$

$$u_h^0 = u_h^0(x, y), \quad v_h^0 = v_h^0(x, y), \quad (33)$$

where $u_h^0(x, y), v_h^0(x, y)$ are some approximations of $\text{Re}\Psi_0(x, y)$ and $\text{Im}\Psi_0(x, y)$, respectively.

Using the idea of split-steps based on the second-order Strang splitting method and combining the analytical solutions (29) and (30) of the nonlinear subproblem and the Crank–Nicolson finite element scheme (31)–(33) for the linear subproblem, we finally propose the fully discrete split-step FEM as follow:

$$u_h^{*,1} = \cos [\lambda |\Psi_h^n|^2 \Delta t / 2] u_h^n - \sin [\lambda |\Psi_h^n|^2 \Delta t / 2] v_h^n, \quad (34)$$

$$v_h^{*,1} = \sin [\lambda |\Psi_h^n|^2 \Delta t / 2] u_h^n + \cos [\lambda |\Psi_h^n|^2 \Delta t / 2] v_h^n, \quad (35)$$

$$\left(\frac{u_h^{*,2} - u_h^{*,1}}{\Delta t}, \chi_1 \right) - \gamma \Lambda_h \left(\frac{v_h^{*,2} + v_h^{*,1}}{2}, \chi_1 \right) = 0, \quad (36)$$

$$\left(\frac{v_h^{*,2} - v_h^{*,1}}{\Delta t}, \chi_2 \right) + \gamma \Lambda_h \left(\frac{u_h^{*,2} + u_h^{*,1}}{2}, \chi_2 \right) = 0, \quad (37)$$

$$u_h^{n+1} = \cos [\lambda |\Psi_h^{*,2}|^2 \Delta t / 2] u_h^{*,2} - \sin [\lambda |\Psi_h^{*,2}|^2 \Delta t / 2] v_h^{*,2}, \quad (38)$$

$$v_h^{n+1} = \sin [\lambda |\Psi_h^{*,2}|^2 \Delta t / 2] u_h^{*,2} + \cos [\lambda |\Psi_h^{*,2}|^2 \Delta t / 2] v_h^{*,2}, \quad (39)$$

$$u_h^0 = u_h^0(x, y), \quad v_h^0 = v_h^0(x, y), \quad (40)$$

for $\chi_1, \chi_2 \in \mathcal{V}_h$, where $|\Psi_h^{*,2}|^2 = |u_h^{*,2}|^2 + |v_h^{*,2}|^2$, $|\Psi_h^n|^2 = |u_h^n|^2 + |v_h^n|^2$ and $n = 0, 1, 2, \dots, N-1$.

Conservation and convergence

In this section, we derive the discrete conservation law and error estimate for the fully discrete split-step FEM (34)–(40). Before proceeding, we define a projection operator P_h by

$$\Lambda_h(u, v_h) = \Lambda_h(P_h u, v_h), \quad \forall v_h \in \mathcal{V}_h.$$

which maps the elements from $H_0^1(\Omega)$ to $\mathcal{V}_h$ and admits³⁰:

$$\|u - P_h u\|_{0,\Omega} \leq Ch^r \|u\|_{r,\Omega}, \quad (41)$$

when $u \in \mathcal{V}_h \cap H^r(\Omega)$ and fulfills necessary condition. Letting C be a general constant independent of u, v , we next analyze the conservation laws and convergent properties.

Lemma 5.1 (Discrete Gronwall Inequality³³) *If the nonnegative sequences $\{G^n\}_{n=0}^N$ and $\{g^n\}_{n=0}^N$ satisfy*

$$G^{n+1} - G^n \leq a\Delta t G^{n+1} + b\Delta t G^n + \Delta t g^{n+1},$$

where a, b are nonnegative numbers and $T = N\Delta t$. Then

$$\max_{1 \leq n \leq N} G^n \leq \left(G^0 + \Delta t \sum_{k=1}^N g^k \right) e^{2(a+b)T},$$

where Δt satisfies $(a+b)\Delta t \leq \frac{N-1}{2N}$, $N > 1$.

In what follows, we turn to the conservation and convergence for the split-step FEM.

Lemma 5.2 ³⁴*If the exact solution $\Psi(t)$ of Eqs. (2)–(4) is regular enough, then the Strang split-step scheme is consistent of order 2, i.e.,*

$$\|T(\Psi^0) - J_t(\Psi^0)\|_{0,\Omega} \leq C\Delta t^3,$$

where $T = e^{-i\Delta t(L+N)}$ and C is independent of Δt .

Lemma 5.3 ³⁵*If $\phi, \psi \in H^2(\Omega)$, then we have*

$$\|J_t(\phi) - J_t(\psi)\|_{0,\Omega} \leq e^{C\Delta t} \|\phi - \psi\|_{0,\Omega}.$$

Theorem 5.1 *The fully discrete split-step finite element scheme (34)–(40) satisfies the discrete mass conservation law, i.e.,*

$$M_h^n = M_h^0, \quad \forall n = 1, 2, \dots, N,$$

where $M_h^n = \|u_h^n\|_{0,\Omega}^2 + \|v_h^n\|_{0,\Omega}^2$.

Proof Letting

$$\partial_t u^* = \frac{u^{*,2} - u^{*,1}}{\Delta t}, \quad \bar{u}^* = \frac{u^{*,1} + u^{*,2}}{2},$$

and in (36)–(37), taking $\chi_1 = \bar{u}_h^*$, $\chi_2 = \bar{v}_h^*$, we have

$$(\bar{\partial}_t u_h^*, \bar{u}_h^*) - \gamma \Lambda_h(\bar{v}_h^*, \bar{u}_h^*) = 0, \quad (42)$$

$$(\bar{\partial}_t v_h^*, \bar{v}_h^*) + \gamma \Lambda_h(\bar{u}_h^*, \bar{v}_h^*) = 0. \quad (43)$$

In view of

$$\begin{aligned}(\bar{\partial}_t u_h^*, \bar{u}_h^*) &= \frac{\|u_h^{*,2}\|_{0,\Omega}^2 - \|u_h^{*,1}\|_{0,\Omega}^2}{2\Delta t}, \\(\bar{\partial}_t v_h^*, \bar{v}_h^*) &= \frac{\|v_h^{*,2}\|_{0,\Omega}^2 - \|v_h^{*,1}\|_{0,\Omega}^2}{2\Delta t},\end{aligned}$$

the sum of Eqs. (42)–(43) leads to

$$(\bar{\partial}_t u_h^*, \bar{u}_h^*) + (\bar{\partial}_t v_h^*, \bar{v}_h^*) = \frac{\|u_h^{*,2}\|_{0,\Omega}^2 - \|u_h^{*,1}\|_{0,\Omega}^2}{2\Delta t} + \frac{\|v_h^{*,2}\|_{0,\Omega}^2 - \|v_h^{*,1}\|_{0,\Omega}^2}{2\Delta t} = 0,$$

which implies $\|u_h^{*,2}\|_{0,\Omega}^2 + \|v_h^{*,2}\|_{0,\Omega}^2 = \|u_h^{*,1}\|_{0,\Omega}^2 + \|v_h^{*,1}\|_{0,\Omega}^2$.

Using triangular function formula, we obtain

$$\begin{aligned}|u_h^{*,1}|^2 + |v_h^{*,1}|^2 &= (\cos [\lambda |\Psi_h^n|^2 \Delta t / 2] u_h^n - \sin [\lambda |\Psi_h^n|^2 \Delta t / 2] v_h^n)^2 \\&\quad + (\sin [\lambda |\Psi_h^n|^2 \Delta t / 2] u_h^n + \cos [\lambda |\Psi_h^n|^2 \Delta t / 2] v_h^n)^2 \\&= (\cos^2 [\lambda |\Psi_h^n|^2 \Delta t / 2] + \sin^2 [\lambda |\Psi_h^n|^2 \Delta t / 2]) |u_h^n|^2 \\&\quad + (\sin^2 [\lambda |\Psi_h^n|^2 \Delta t / 2] + \cos^2 [\lambda |\Psi_h^n|^2 \Delta t / 2]) |v_h^n|^2 = |u_h^n|^2 + |v_h^n|^2,\end{aligned}$$

Similarly, we can prove $|u_h^{n+1}|^2 + |v_h^{n+1}|^2 = |u_h^{*,2}|^2 + |v_h^{*,2}|^2$, and then $\|u_h^{n+1}\|_{0,\Omega}^2 + \|v_h^{n+1}\|_{0,\Omega}^2 = \|u_h^n\|_{0,\Omega}^2 + \|v_h^n\|_{0,\Omega}^2$. This leads to $M_h^n = M_h^{n-1}$, thus $M_h^n = M_h^{n-1} = \dots = M_h^0$.

Theorem 5.2 Suppose that u^n, v^n are the exact solutions to the linear subproblem (21)–(22) at t_n and u_h^n, v_h^n are obtained by the scheme (31)–(33). Then

$$\max_{1 \leq n \leq N} \{\|e_u^n\|_{0,\Omega}, \|e_v^n\|_{0,\Omega}\} \leq C(u, v, T)(\Delta t^2 + h^r), \quad (44)$$

where $e_u^n = u_h^n - u^n$, $e_v^n = v_h^n - v^n$, $u_h^0 = P_h \text{Re} \Psi_0(x, y)$, $v_h^0 = P_h \text{Im} \Psi_0(x, y)$ and $C(u, v, T)$ is a constant related to u, v, T but independent of Δt and h .

Proof Split the error into two parts, i.e., $e_v^n = \eta_v^n + \rho_v^n$, with $\eta_v^n = v_h^n - P_h v^n$, $\rho_v^n = P_h v^n - v^n$ and $v = u, v$. Taking $t_{n+\frac{1}{2}} = t_n + \Delta t/2$ and comparing Eq. (21) with Eq. (31), we have

$$(\bar{\partial}_t e_u^{n+1}, \chi_1) - \gamma A_h(\bar{e}_v^{n+\frac{1}{2}}, \chi_1) = (\partial_t u(t_{n+\frac{1}{2}}) - \bar{\partial}_t u^{n+1}, \chi_1) + \gamma A_h(\bar{v}^{n+\frac{1}{2}} - v(t_{n+\frac{1}{2}}), \chi_1).$$

In view of $A_h(\rho_v^n, \chi_1) = 0$, there exists

$$\begin{aligned}(\bar{\partial}_t \eta_u^{n+1}, \chi_1) - \gamma A_h(\bar{\eta}_v^{n+\frac{1}{2}}, \chi_1) &= (\partial_t u(t_{n+\frac{1}{2}}) - \bar{\partial}_t u^{n+1}, \chi_1) \\&\quad + (\bar{\partial}_t u^{n+1} - \bar{\partial}_t P_h u^{n+1}, \chi_1) + \gamma A_h(\bar{v}^{n+\frac{1}{2}} - v(t_{n+\frac{1}{2}}), \chi_1).\end{aligned} \quad (45)$$

Similarly, comparing Eq. (22) with Eq. (32), we have

$$\begin{aligned}(\bar{\partial}_t \eta_v^{n+1}, \chi_2) + \gamma A_h(\bar{\eta}_u^{n+\frac{1}{2}}, \chi_2) &= (\partial_t v(t_{n+\frac{1}{2}}) - \bar{\partial}_t v^{n+1}, \chi_2) \\&\quad + (\bar{\partial}_t v^{n+1} - \bar{\partial}_t P_h v^{n+1}, \chi_2) + \gamma A_h(u(t_{n+\frac{1}{2}}) - \bar{u}^{n+\frac{1}{2}}, \chi_2).\end{aligned} \quad (46)$$

Taking $\chi_1 = \bar{\eta}_u^{n+\frac{1}{2}}$ in Eq. (45) and $\chi_2 = \bar{\eta}_v^{n+\frac{1}{2}}$ in Eq. (46), respectively, and summing them together, we obtain

$$\begin{aligned}(\bar{\partial}_t \eta_u^{n+1}, \bar{\eta}_u^{n+\frac{1}{2}}) + (\bar{\partial}_t \eta_v^{n+1}, \bar{\eta}_v^{n+\frac{1}{2}}) &= (\omega_u^1 + \omega_u^2, \bar{\eta}_u^{n+\frac{1}{2}}) + \gamma A_h(u(t_{n+\frac{1}{2}}) - \bar{u}^{n+\frac{1}{2}}, \bar{\eta}_v^{n+\frac{1}{2}}) \\&\quad + (\omega_v^1 + \omega_v^2, \bar{\eta}_v^{n+\frac{1}{2}}) + \gamma A_h(\bar{v}^{n+\frac{1}{2}} - v(t_{n+\frac{1}{2}}), \bar{\eta}_u^{n+\frac{1}{2}}),\end{aligned} \quad (47)$$

where

$$\omega_u^1 = \partial_t v(t_{n+\frac{1}{2}}) - \bar{\partial}_t v^{n+1}, \quad \omega_u^2 = \bar{\partial}_t v^{n+1} - \bar{\partial}_t P_h v^{n+1}.$$

For the terms on the left side of Eq. (47), we have

$$(\bar{\partial}_t \eta_v^{n+1}, \bar{\eta}_v^{n+\frac{1}{2}}) = \frac{\|\eta_v^{n+1}\|_{0,\Omega}^2 - \|\eta_v^n\|_{0,\Omega}^2}{2\Delta t}, \quad (48)$$

while for the terms on the right side, we can deduce that

$$\begin{aligned}
 (\omega_v^1 + \omega_v^2, \bar{\eta}_v^{n+\frac{1}{2}}) &\leq \|\omega_v^1 + \omega_v^2\|_{0,\Omega} \|\bar{\eta}_v^{n+\frac{1}{2}}\|_{0,\Omega} \\
 &\leq \|\omega_v^1\|_{0,\Omega} \|\bar{\eta}_v^{n+\frac{1}{2}}\|_{0,\Omega} + \|\omega_v^2\|_{0,\Omega} \|\bar{\eta}_v^{n+\frac{1}{2}}\|_{0,\Omega} \\
 &\leq \|\omega_v^1\|_{0,\Omega} \|\eta_v^{n+1}\|_{0,\Omega} + \|\omega_v^2\|_{0,\Omega} \|\eta_v^{n+1}\|_{0,\Omega} \\
 &\quad + \|\omega_v^1\|_{0,\Omega} \|\eta_v^n\|_{0,\Omega} + \|\omega_v^2\|_{0,\Omega} \|\eta_v^n\|_{0,\Omega} \\
 &\leq \|\omega_v^1\|_{0,\Omega}^2 + \|\omega_v^2\|_{0,\Omega}^2 + \|\eta_v^{n+1}\|_{0,\Omega}^2 + \|\eta_v^n\|_{0,\Omega}^2.
 \end{aligned} \tag{49}$$

From Lemma 2.3 and Taylor formula, it follows that

$$\begin{aligned}
 \Lambda_h(u(t_{n+\frac{1}{2}}) - \bar{u}^{n+\frac{1}{2}}, \bar{\eta}_v^{n+\frac{1}{2}}) &= \Lambda_h(u(t_{n+\frac{1}{2}}) - \bar{u}^{n+\frac{1}{2}}, \bar{\eta}_v^{n+\frac{1}{2}}) \\
 &= \left(\frac{\partial^{2\alpha}(u(t_{n+\frac{1}{2}}) - \bar{u}^{n+\frac{1}{2}})}{\partial |x|^{2\alpha}}, \bar{\eta}_v^{n+\frac{1}{2}} \right) \\
 &\quad + \left(\frac{\partial^{2\alpha}(u(t_{n+\frac{1}{2}}) - \bar{u}^{n+\frac{1}{2}})}{\partial |y|^{2\alpha}}, \bar{\eta}_v^{n+\frac{1}{2}} \right) \\
 &\leq \left\| \frac{\partial^{2\alpha}(u(t_{n+\frac{1}{2}}) - \bar{u}^{n+\frac{1}{2}})}{\partial |x|^{2\alpha}} \right\|_{0,\Omega}^2 \\
 &\quad + \left\| \frac{\partial^{2\alpha}(u(t_{n+\frac{1}{2}}) - \bar{u}^{n+\frac{1}{2}})}{\partial |y|^{2\alpha}} \right\|_{0,\Omega}^2 + \|\eta_v^{n+1}\|_{0,\Omega}^2 + \|\eta_v^n\|_{0,\Omega}^2,
 \end{aligned} \tag{50}$$

and similarly, we can obtain

$$\begin{aligned}
 \Lambda_h(\bar{v}^{n+\frac{1}{2}} - v(t_{n+\frac{1}{2}}), \bar{\eta}_u^{n+\frac{1}{2}}) &\leq \left\| \frac{\partial^{2\alpha}(\bar{v}^{n+\frac{1}{2}} - v(t_{n+\frac{1}{2}}))}{\partial |x|^{2\alpha}} \right\|_{0,\Omega}^2 \\
 &\quad + \left\| \frac{\partial^{2\alpha}(\bar{v}^{n+\frac{1}{2}} - v(t_{n+\frac{1}{2}}))}{\partial |y|^{2\alpha}} \right\|_{0,\Omega}^2 + \|\eta_u^{n+1}\|_{0,\Omega}^2 + \|\eta_u^n\|_{0,\Omega}^2.
 \end{aligned} \tag{51}$$

On the other hand, by the property (41) and Taylor formula, there exist

$$\begin{aligned}
 \|\omega_v^1\|_{0,\Omega}^2 &= \|\partial_t v(t_{n+\frac{1}{2}}) - \bar{\partial}_t v^{n+1}\|_{0,\Omega}^2 \\
 &= \left\| \partial_t v(t_{n+\frac{1}{2}}) - \frac{v(t_{n+1}) - v(t_{n+\frac{1}{2}}) + v(t_{n+\frac{1}{2}}) - v(t_n)}{\Delta t} \right\|_{0,\Omega}^2 \\
 &= \frac{1}{4\Delta t^2} \left\| \int_{t_n}^{t_{n+\frac{1}{2}}} (t - t_n)^2 v_{ttt} dt + \int_{t_{n+\frac{1}{2}}}^{t_{n+1}} (t_{n+1} - t)^2 v_{ttt} dt \right\|_{0,\Omega}^2 \\
 &\leq C\Delta t^3 \int_{t_n}^{t_{n+1}} \|v_{ttt}\|_{0,\Omega}^2 dt,
 \end{aligned} \tag{52}$$

and

$$\|\omega_v^2\|_{0,\Omega}^2 = \|\bar{\partial}_t v^{n+1} - \bar{\partial}_t P_h v^{n+1}\|_{0,\Omega}^2 \leq \frac{1}{\Delta t} \int_{t_n}^{t_{n+1}} \|v_t - P_h v_t\|_{0,\Omega}^2 dt \leq \frac{Ch^{2r}}{\Delta t} \int_{t_n}^{t_{n+1}} \|v_t\|_{r,\Omega}^2 dt, \tag{53}$$

while for the terms related to the fractional derivatives, there holds

$$\begin{aligned}
 \left\| \frac{\partial^{2\alpha}(v(t_{n+\frac{1}{2}}) - \bar{v}^{n+\frac{1}{2}})}{\partial |\xi|^{2\alpha}} \right\|_{0,\Omega}^2 &= \frac{1}{4} \left\| \frac{\partial^{2\alpha}(v(t_{n+\frac{1}{2}}) - v(t_n))}{\partial |\xi|^{2\alpha}} + \frac{\partial^{2\alpha}(v(t_{n+\frac{1}{2}}) - v(t_{n+1}))}{\partial |\xi|^{2\alpha}} \right\|_{0,\Omega}^2 \\
 &= \frac{1}{4} \left\| \int_{t_n}^{t_{n+\frac{1}{2}}} (t - t_n) \frac{\partial^{2\alpha} v_{tt}}{\partial |\xi|^{2\alpha}} dt + \int_{t_{n+\frac{1}{2}}}^{t_{n+1}} (t_{n+1} - t) \frac{\partial^{2\alpha} v_{tt}}{\partial |\xi|^{2\alpha}} dt \right\|_{0,\Omega}^2 \\
 &\leq C\Delta t^3 \int_{t_n}^{t_{n+1}} \left\| \frac{\partial^{2\alpha} v_{tt}}{\partial |\xi|^{2\alpha}} \right\|_{0,\Omega}^2 dt \\
 &\leq C\Delta t^3 \int_{t_n}^{t_{n+1}} \|v_{tt}\|_{2\alpha,\Omega}^2 dt,
 \end{aligned} \tag{54}$$

with $\xi = x$ or y . Substituting (48)–(54) into (47), it suffices to show that

$$\begin{aligned} \|\eta_u^{n+1}\|_{0,\Omega}^2 + \|\eta_v^{n+1}\|_{0,\Omega}^2 - \|\eta_u^n\|_{0,\Omega}^2 - \|\eta_v^n\|_{0,\Omega}^2 &\leq 4\Delta t (\|\eta_u^{n+1}\|_{0,\Omega}^2 + \|\eta_v^{n+1}\|_{0,\Omega}^2) \\ &\quad + 4\Delta t (\|\eta_u^n\|_{0,\Omega}^2 + \|\eta_v^n\|_{0,\Omega}^2) \\ &\quad + \Delta t H^{n-1}(u) + \Delta t H^{n-1}(v), \end{aligned} \quad (55)$$

where

$$H^{n-1}(v) = C\Delta t^3 \int_{t_n}^{t_{n+1}} \|v_{ttt}\|_{0,\Omega}^2 dt + \frac{Ch^{2r}}{\Delta t} \int_{t_n}^{t_{n+1}} \|v_t\|_{r,\Omega}^2 dt + C\Delta t^3 \int_{t_n}^{t_{n+1}} \|v_{tt}\|_{2\alpha,\Omega}^2 dt.$$

Apply the discrete Gronwall inequality to (55) and note $\|\eta_u^0\|_{0,\Omega}^2 = \|\eta_v^0\|_{0,\Omega}^2 = 0$. Then

$$\begin{aligned} \max_{1 \leq n \leq N} \{\|\eta_u^{n+1}\|_{0,\Omega}^2 + \|\eta_v^{n+1}\|_{0,\Omega}^2\} &\leq \left[\Delta t \sum_{k=1}^N (H^{k-1}(u) + H^{k-1}(v)) \right] e^{16T}, \\ &\leq C\Delta t^4 \int_0^T \|u_{ttt}\|_{r,\Omega}^2 + \|v_{ttt}\|_{r,\Omega}^2 dt + Ch^{2r} \int_0^T \|u_t\|_{r,\Omega}^2 \\ &\quad + \|v_t\|_{r,\Omega}^2 dt + C\Delta t^4 \int_0^T \|u_{tt}\|_{2\alpha,\Omega}^2 + \|v_{tt}\|_{2\alpha,\Omega}^2 dt, \end{aligned}$$

which implies

$$\max_{1 \leq n \leq N} \{\|\eta_u^{n+1}\|_{0,\Omega}, \|\eta_v^{n+1}\|_{0,\Omega}\} \leq \bar{C}(u, v, T)(\Delta t^2 + h^r). \quad (56)$$

with $\bar{C}(u, v, T)$ being a constant only related to u, v and T . Using the property (41), we obtain

$$\begin{aligned} \max_{1 \leq n \leq N} \{\|e_u^n\|_{0,\Omega}, \|e_v^n\|_{0,\Omega}\} &\leq \max_{1 \leq n \leq N} \{\|\rho_u^n\|_{0,\Omega} + \|\eta_u^n\|_{0,\Omega}, \|\rho_v^n\|_{0,\Omega} + \|\eta_v^n\|_{0,\Omega}\} \\ &\leq Ch^r \max_{1 \leq n \leq N} \{\|u^n\|_{r,\Omega}, \|v^n\|_{r,\Omega}\} + \bar{C}(u, v, T)(\Delta t^2 + h^r) \\ &\leq C(u, v, T)(\Delta t^2 + h^r), \end{aligned}$$

and the proof is finally completed. $\square$

Theorem 5.3 Suppose that $u(t_n), v(t_n)$ are the exact solutions to Eqs. (5)–(9) at t_n and u_h^n, v_h^n are the solutions obtained by the split-step FEM (34)–(40). Then

$$\max_{1 \leq n \leq N} \{\|e_u^n\|_{0,\Omega}, \|e_v^n\|_{0,\Omega}\} \leq C(u, v, T)(\Delta t^2 + h^r), \quad (57)$$

where $e_u^n = u_h^n - u(t_n)$, $e_v^n = v_h^n - v(t_n)$, $u_h^0 = P_h \operatorname{Re} \Psi_0(x, y)$, $v_h^0 = P_h \operatorname{Im} \Psi_0(x, y)$ and $C(u, v, T)$ is a constant related to u, v, T but independent of Δt and h .

Proof Let $\Psi^n = u^n + iv^n$ be the split-step solutions yielded by the scheme (18). Using the Lady Windermere's fan, we obtain

$$\Psi(t_n) - \Psi^n = \sum_{k=1}^n (J_t^{n-k}) [J_t T^{k-1}(\Psi^0) - T^k(\Psi^0)].$$

From Lemma 5.2, 5.3, it follows that

$$\begin{aligned} \|\Psi(t_n) - \Psi^n\|_{0,\Omega} &\leq \sum_{k=1}^n \|(J_t^{n-k}) [J_t T^{k-1}(\Psi^0) - T^k(\Psi^0)]\|_{0,\Omega} \\ &\leq \sum_{k=1}^n e^{C(t_n - t_k)} \|J_t T^{k-1}(\Psi^0) - T^k(\Psi^0)\|_{0,\Omega} \\ &\leq \sum_{k=1}^n e^{C(n-k)\Delta t} \tilde{C} \Delta t^3 \leq C \Delta t^2, \end{aligned}$$

with $n = 1, 2, \dots, N$, which implies

$$\max_{1 \leq n \leq N} \{\|u(t_n) - u^n\|_{0,\Omega}, \|v(t_n) - v^n\|_{0,\Omega}\} \leq C \Delta t^2, \quad (58)$$

where $\tilde{C}$ is also unrelated to Δt .

On the other hand, since the solutions to the nonlinear problem are integrated exactly in physical space, according to the property of the standard Strang split-step technique and Theorem 5.2, it is trivial to validate that

$$\max_{1 \leq n \leq N} \{ \|u_h^n - u^n\|_{0,\Omega}, \|v_h^n - v^n\|_{0,\Omega} \} \leq C(\Delta t^2 + h^r), \quad (59)$$

and thus by (58) and (59), we can easily obtain the above results. $\square$

Illustrative examples

In this section, we perform some numerical examples to confirm the theoretical results and illustrate the capability of our split-step FEM, including both one- and two-dimensional cases. We discuss the convergent accuracy of this method by constructive example and simulate the dynamics of the wave solutions governed by the Riesz space-fractional NLS in both one- and two-dimensions, where the mass conservative property will be testified. The convergent rate of the real and imaginary errors in time and space is calculated by

$$Rate = \begin{cases} \log_2 \left(\frac{error_{\Delta t_1}}{error_{\Delta t_2}} \right) / \log_2 \left(\frac{\Delta t_1}{\Delta t_2} \right), & \text{in time,} \\ \log_2 \left(\frac{error_{h_1}}{error_{h_2}} \right) / \log_2 \left(\frac{h_1}{h_2} \right), & \text{in space,} \end{cases}$$

where $error_{\Delta t_k}$, $error_{h_k}$ are the numerical errors in time and space corresponding to the time stepsize Δt_k and spatial meshsize h_k with $k = 1, 2$. To show its computational efficiency, we present the following two schemes:

- Crank–Nicolson FEM with Newton's iteration:

$$\begin{aligned} (\bar{\partial}_t u_h^{n+1}, \chi_1) - \gamma \Lambda_h(\bar{v}_h^{n+\frac{1}{2}}, \chi_1) + \lambda(|\bar{\Psi}_h^{n+\frac{1}{2}}|^2 \bar{v}_h^{n+\frac{1}{2}}, \chi_1) &= 0, \\ (\bar{\partial}_t v_h^{n+1}, \chi_2) + \gamma \Lambda_h(\bar{u}_h^{n+\frac{1}{2}}, \chi_2) - \lambda(|\bar{\Psi}_h^{n+\frac{1}{2}}|^2 \bar{u}_h^{n+\frac{1}{2}}, \chi_2) &= 0, \end{aligned}$$

where $|\bar{\Psi}_h^{n+\frac{1}{2}}|^2 = |\bar{u}_h^{n+\frac{1}{2}}|^2 + |\bar{v}_h^{n+\frac{1}{2}}|^2$,

- Linearized Crank–Nicolson FEM³⁰:

$$\begin{aligned} (\bar{\partial}_t u_h^{n+1}, \chi_1) - \gamma \Lambda_h(\bar{v}_h^{n+\frac{1}{2}}, \chi_1) + \lambda(|\hat{\Psi}_h^n|^2 \bar{v}_h^{n+\frac{1}{2}}, \chi_1) &= 0, \\ (\bar{\partial}_t v_h^{n+1}, \chi_2) + \gamma \Lambda_h(\bar{u}_h^{n+\frac{1}{2}}, \chi_2) - \lambda(|\hat{\Psi}_h^n|^2 \bar{u}_h^{n+\frac{1}{2}}, \chi_2) &= 0, \end{aligned}$$

and to start it, u_h^1, v_h^1 are computed by the below explicit-implicit scheme:

$$\begin{aligned} (\bar{\partial}_t u_h^1, \chi_1) - \gamma \Lambda_h(v_h^1, \chi_1) + \lambda(|\Psi_h^0|^2 v_h^1, \chi_1) &= 0, \\ (\bar{\partial}_t v_h^1, \chi_2) + \gamma \Lambda_h(u_h^1, \chi_2) - \lambda(|\Psi_h^0|^2 u_h^1, \chi_2) &= 0, \end{aligned}$$

where $|\hat{\Psi}_h^n|^2 = |\hat{u}_h^n|^2 + |\hat{v}_h^n|^2$, $|\Psi_h^0|^2 = |u_h^0|^2 + |v_h^0|^2$, $\hat{u}_h^n = \frac{3u_h^n - u_h^{n-1}}{2}$, and $\hat{v}_h^n = \frac{3v_h^n - v_h^{n-1}}{2}$.

In all tests, we employ linear interpolation, i.e., $r = 2$, then by the theoretical results, we expect that the errors of real and imaginary parts would steadily decline with the desired convergent rate both in time and space. In addition, all the tests will be done by MATLAB R2015b on a personal PC and the last one is implemented by the PC configuration: WIN 7 Ultimate, AMD Athlon(tm) II $\times$ 4 640 3.00 GHz CPU and 8 GB DDR3 RAM.

One-dimensional problem

Example 1 Consider the semiclassical space-fractional NLS³⁶:

$$i\varepsilon \frac{\partial \Psi(x, t)}{\partial t} + \varepsilon^{2\alpha} \frac{\partial^{2\alpha} \Psi(x, t)}{\partial |x|^{2\alpha}} + \lambda |\Psi(x, t)|^2 \Psi(x, t) = 0$$

on $\Omega = [-\ell, \ell]$ with two cases of initial conditions.

In the first case, letting $\varepsilon = 1$ and $\lambda = 2$, we simulate the propagation of a single soliton over time subjected to the initial value of Gaussians type:

$$\Psi(x, 0) = e^{2ix} \operatorname{sech}(x), \quad x \in \mathbb{R},$$

and if $\alpha = 1$, the soliton solution is

$$\Psi(x, t) = e^{i(2x-3t)} \operatorname{sech}(x-4t).$$

When $\ell \gg 0$, the exact solution can be negligible outside Ω because $\Psi(x, 0)$ polynomially or exponentially decays to zero. Therefore, we enforce

$$\Psi(x, t) = 0, \quad (x; t) \in \mathbb{R} \setminus \Omega \times (0, T].$$

Letting $\Delta t = 1.0e-02$, $h = 1/20$ and $\ell = 30$, we plot the evolution of single soliton for different α in Fig. 1. As we observed, the fractional power α can affect shape of the solitons. More precisely, when α becomes smaller, its amplitude $|\Psi_h^n(x)|$ become higher and the wave travels more slowly. Reletting $\Delta t = 1.0e-03$ and $\ell = 10$, we test the convergent accuracy of split-step FEM for $\alpha = 1$ and show the numerical results at $t = 1$ in Table 1. It is seen that our method converges to the exact solution with theoretical convergent rates.

In the second case, we simulate the dynamics of defocusing NLS with $\varepsilon = 0.1$, $\lambda = -1$ and the initial value

$$\Psi(x, 0) = \operatorname{sech}(x), \quad x \in \mathbb{R}.$$

Letting $\Delta t = 5.0e-03$, $h = 1/20$ and $\ell = 10$, we enforce $\Psi(x, t) = 0$ with $x \in \mathbb{R} \setminus \Omega$ as the first case and present the defocusing solutions for different α in Fig. 2. As we observed, the nonlinear interactions are repulsive instead of attractive. At some point, the hump splits into two smaller ones propagating to two different directions. When α becomes smaller, such phenomena appears more manifest in a finite time period. Reletting $\Delta t = 0.1$ and $\ell = 20$, the values of discrete mass M_h^n at different time points for $\alpha = 0.7, 0.8$ and 0.9 are tabulated in Table 2. As we see, the discrete mass conservative law is preserved by this split-step FEM.

Two-dimensional problems

Example 2 Consider the fractional NLS type equation

$$i \frac{\partial \Psi(x, y, t)}{\partial t} + \frac{\partial^{2\alpha} \Psi(x, y, t)}{\partial |x|^{2\alpha}} + \frac{\partial^{2\alpha} \Psi(x, y, t)}{\partial |y|^{2\alpha}} + \lambda |\Psi(x, y, t)|^2 \Psi(x, y, t) = R(x, y, t),$$

on $\Omega = [0, 1]^2$, subjected to

$$\begin{aligned} \Psi(x, y, 0) &= x^2(1-x)^2y^2(1-y)^2, \quad (x, y) \in \Omega, \\ \Psi(x, y, t) &= 0, \quad (x, y; t) \in \mathbb{R}^2 \setminus \Omega \times (0, T]. \end{aligned}$$

and the right-hand function $R(x, y, t)$ is selected to enforce the analytical solution

$$\Psi(x, y, t) = t^5 e^{ist} x^2(1-x)^2y^2(1-y)^2.$$

We test the convergent accuracy of the proposed split-step FEM. Letting $\lambda = 0$, the governing equation degenerates to the linear problem (21) and (22) and the scheme is actually the Crank-Nicolson finite element scheme (31)–(33). Taking $s = 1$ and $h = 1/35$, we firstly compute the results at $t = 0.5$ by using different time stepsizes and the numerical results for $\alpha = 0.8, 0.9$ are tabulated in Table 3. Letting $\Delta t = 2.0e-04$, we compute the results by using different spatial meshsizes and present them in Table 4. Further, retaking $\lambda = 1$ and $s = 0$, we test the accuracy of the split-step FEM and tabulate the numerical results for $\alpha = 0.85, 0.95$ in time and space in Tables 5 and 6, respectively. From these tables, it can be seen that both schemes nearly achieve the convergent rate 2, which are in line with our expectations.

Example 3 Consider the space-fractional NLS¹⁰

$$i \frac{\partial \Psi(x, y, t)}{\partial t} + \frac{\partial^{2\alpha} \Psi(x, y, t)}{\partial |x|^{2\alpha}} + \frac{\partial^{2\alpha} \Psi(x, y, t)}{\partial |y|^{2\alpha}} + \lambda |\Psi(x, y, t)|^2 \Psi(x, y, t) = 0,$$

on $\Omega = [-\ell, \ell]^2$, subjected to

$$\Psi(x, y, 0) = \frac{2}{\sqrt{\pi}} \exp(-x^2 - y^2), \quad (x, y) \in \mathbb{R}^2.$$

When $\ell \gg 0$, the wave function can also be negligible outside Ω because $\Psi(x, y, 0)$ exponentially decays to zero. Therefore, we enforce the homogeneous boundary condition:

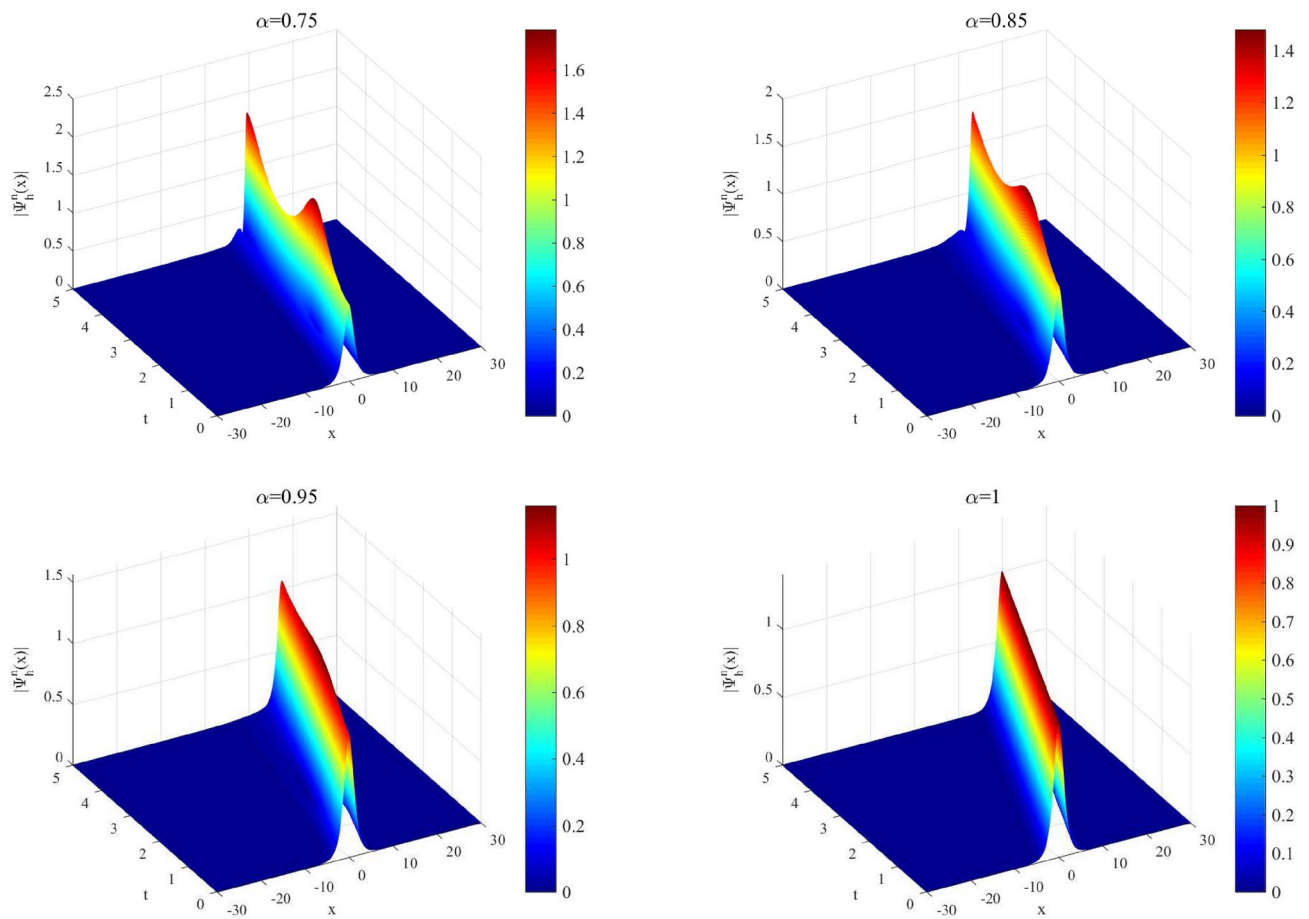


Fig. 1. The evolution of discrete mass M_h^n for $\lambda = 0$ (left) and $\lambda = 1$ (right).

h	$\ e_u\ _{0,\Omega}$	Rate	$\ e_v\ _{0,\Omega}$	Rate
1/5	1.3608e-01	–	1.3890e-01	–
1/10	3.4829e-02	1.97	3.5661e-02	1.96
1/20	8.9829e-03	1.96	9.3381e-03	1.93

Table 1. Numerical errors of the split-step FEM at $t = 1$ with $\Delta t = 1.0e - 03$.

$$\Psi(x, y, t) = 0, \quad (x, y; t) \in \mathbb{R}^2 \setminus \Omega \times (0, T].$$

In order to examine the discrete mass conservative law, we take $\Delta t = 0.1$, $h = 0.6$ and a large enough $\ell = 8$. Consider the cases of $\lambda = 0$ and $\lambda = 1$. In Fig. 3, the evolution of the value of discrete mass M_h^n in the time interval $[0, 10]$ with $\alpha = 0.65$ for both cases is presented, respectively. From the figures, we conclude that the discrete mass is preserved by our methods during the whole computing process and the proposed schemes can be applied to long-time simulation.

Next, we simulate the dynamic of the wave solution governed by the Riesz space-fractional NLS via the split-step FEM and discuss how the fractional power α affects the dynamic of wave solution. For this purpose, letting $\lambda = 1$, $\ell = 5$, $\Delta t = 0.01$ and $h = 1/3$, we display the profile of the modulus of the wave solutions $|\Psi_h^n(x, y)|$ for $\alpha = 0.7$ at the times $t = 0, 0.3, 0.6$ and 0.9 in Fig. 4. It is observed that as time goes by, the altitude of wave solution gets smaller and smaller, while its width gets bigger and bigger, which coincide with the variation tendency of the classical wave. Meanwhile, we plot the profile and contour map of the modulus of wave solutions at $t = 0.5$ for $\alpha = 0.55, 0.75, 0.95$ and all of them are presented in Figs. 5, 6 and 7, respectively. For comparison, we also plot the profile and contour map of the modulus of wave solution in the case of $\alpha = 1$, which are given in Fig. 8. From these figures, we easily observe that the fractional power significantly affects the shape of wave solution. More precisely, as α decreases, the wave solution decays faster and the shape of the wave becomes higher and more steep than those with larger α . When $\alpha = 1$, the wave solution collapses to the one governed by the classical NLS.

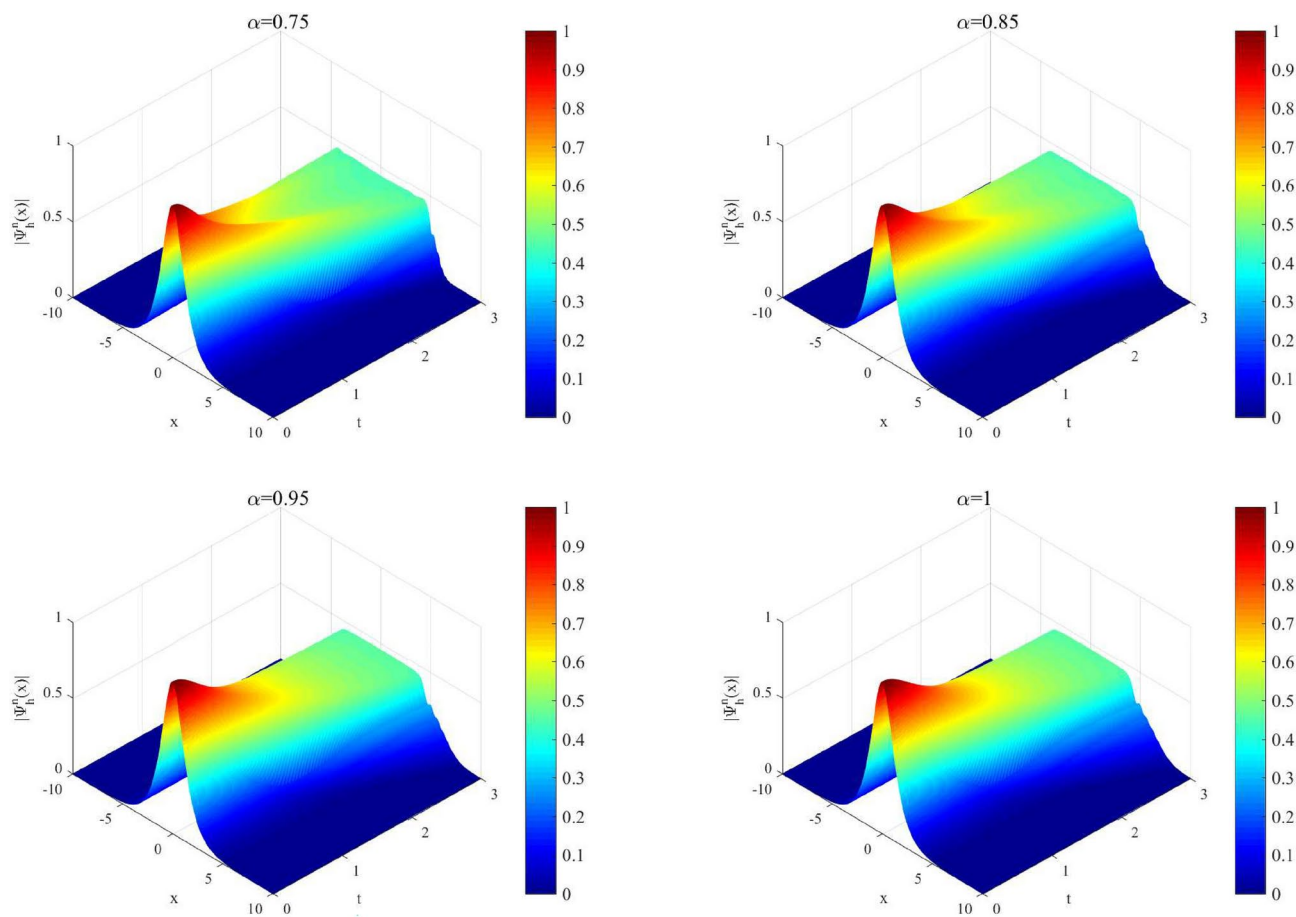


Fig. 2. The evolution of the wave solution at the time points $t = 0, 0.3, 0.6$ and 0.9 .

	$\alpha = 0.7$	$\alpha = 0.8$	$\alpha = 0.9$
$t = 0$	2.0000000000000001	2.0000000000000001	2.0000000000000001
$t = 2$	2.000000000134017	2.000000000008692	2.00000000000656
$t = 4$	2.000000000331438	2.00000000019515	2.00000000000226
$t = 6$	2.000000000526932	2.00000000034813	2.00000000000503
$t = 8$	2.000000001662211	2.00000000045921	2.00000000001053
$t = 10$	2.000000003200945	2.00000000050882	2.00000000001981

Table 2. The values of discrete mass M_h^n for $\alpha = 0.7, 0.8$ and 0.9 at different t .

Δt	$\alpha = 0.8$				$\alpha = 0.9$			
	$\ e_u\ _{0,\Omega}$	Rate	$\ e_v\ _{0,\Omega}$	Rate	$\ e_u\ _{0,\Omega}$	Rate	$\ e_v\ _{0,\Omega}$	Rate
1/4	1.3492e-05	–	1.8760e-05	–	1.5446e-05	–	1.8931e-05	–
1/8	3.3610e-06	2.01	5.8009e-06	1.69	4.0101e-06	1.95	5.8518e-06	1.69
1/16	8.8615e-07	1.92	1.6163e-06	1.84	1.0603e-06	1.92	1.5997e-06	1.87

Table 3. Numerical errors at $t = 0.5$ in time direction with $h = 1/35$ for $\lambda = 0$.

At last, to show the advantage in computational efficiency, we compare the CPU times of our split-step FEM, the pure Crank-Nicolson FEM with Newton’s iteration and the linearized Crank-Nicolson FEM as in³⁰ with extrapolation technique. Taking $\alpha = 0.75$, $\ell = 5$ and $T = 0.5$, we tabulate the comparative results in Table 7, where the iteration algorithm is terminated by reaching a solution with the tolerant error $1.0e - 12$. It is observed that the CPU times of our method are less than those of the linearized Crank-Nicolson FEM and much

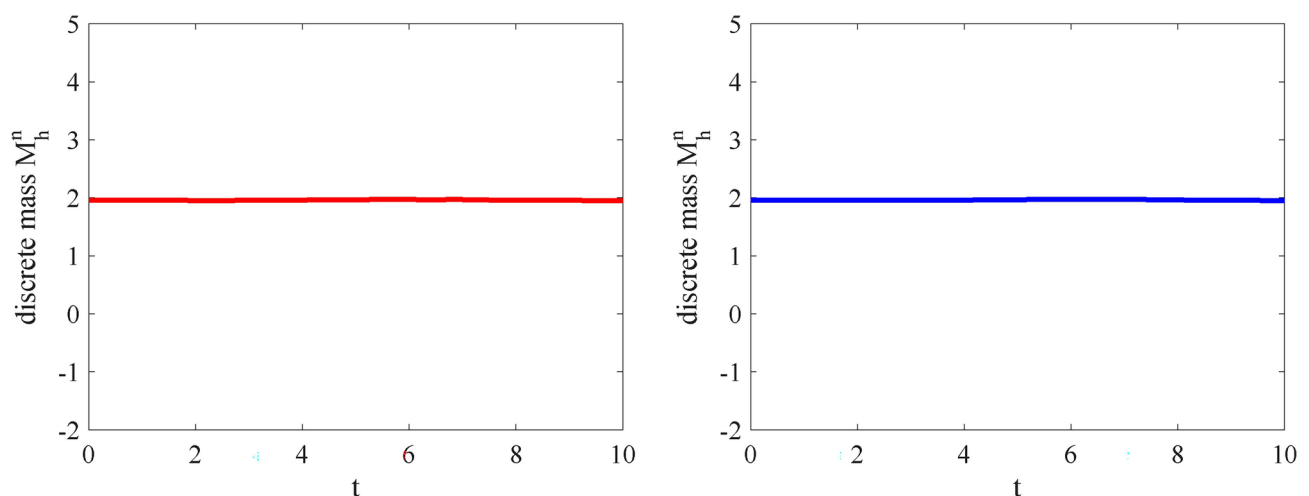
h	$\alpha = 0.8$				$\alpha = 0.9$			
	$\ e_u\ _{0,\Omega}$	Rate	$\ e_v\ _{0,\Omega}$	Rate	$\ e_u\ _{0,\Omega}$	Rate	$\ e_v\ _{0,\Omega}$	Rate
1/5	4.8177e-06	–	5.8883e-06	–	5.3680e-06	–	6.0798e-06	–
1/10	1.2246e-06	1.98	1.5138e-06	1.96	1.3627e-06	1.98	1.5616e-06	1.96
1/20	3.1451e-07	1.96	4.0251e-07	1.91	3.3456e-07	2.03	3.8652e-07	2.01

Table 4. Numerical errors at $t = 0.5$ in space direction with $\Delta t = 2.0e - 04$ for $\lambda = 0$.

Δt	$\alpha = 0.85$				$\alpha = 0.95$			
	$\ e_u\ _{0,\Omega}$	Rate	$\ e_v\ _{0,\Omega}$	Rate	$\ e_u\ _{0,\Omega}$	Rate	$\ e_v\ _{0,\Omega}$	Rate
1/4	2.1993e-05	–	7.1209e-06	–	2.3573e-05	–	6.0014e-06	–
1/8	6.6032e-06	1.74	2.4192e-06	1.56	7.1238e-06	1.73	1.8461e-06	1.70
1/16	1.8105e-06	1.87	6.9523e-07	1.80	1.9158e-06	1.89	4.9376e-07	1.90

Table 5. Numerical errors at $t = 0.5$ in time direction with $h = 1/35$ for $\lambda = 1$.

h	$\alpha = 0.85$				$\alpha = 0.95$			
	$\ e_u\ _{0,\Omega}$	Rate	$\ e_v\ _{0,\Omega}$	Rate	$\ e_u\ _{0,\Omega}$	Rate	$\ e_v\ _{0,\Omega}$	Rate
1/5	6.9333e-06	–	4.0490e-06	–	8.0050e-06	–	3.6425e-06	–
1/10	1.7530e-06	1.98	1.0347e-06	1.97	2.0852e-06	1.94	1.0001e-06	1.86
1/20	4.4017e-07	1.99	2.6543e-07	1.96	5.1489e-07	2.02	2.4952e-07	2.00

Table 6. Numerical errors at $t = 0.5$ in space direction with $\Delta t = 2.0e - 04$ for $\lambda = 1$.**Fig. 3.** The profile (left) and contour map (right) of the modulus of wave solutions for $\alpha = 0.55$.

less than those of the pure Crank-Nicolson FEM with Newton's iteration. The efficiency performance is more excellent with smaller Δt , which demonstrates that this split-step FEM is the most efficient among the above three methods and can reduce the computational cost in practice (Figs. 1, 2).

Concluding remarks

The issues of concern for the high-dimensional Riesz space-fractional NLS is not only the high complexity of the fractional derivatives in space, but also its nonlinearity. The split-step FEM proposed in our work splits the governing equation into the linear and nonlinear subproblems, which avoids the inner iteration at each time layer, thereby reducing the computational complexity. The mass conservation law has been well inherited, which means that it can keep the phase and shape of the waves governed by the space-fractional NLS for long time. The analysis of the unconditional convergence for this fully discrete split-step FEM is also presented. The numerical results and simulation of wave dynamics confirm its excellent actual performance and conservative property

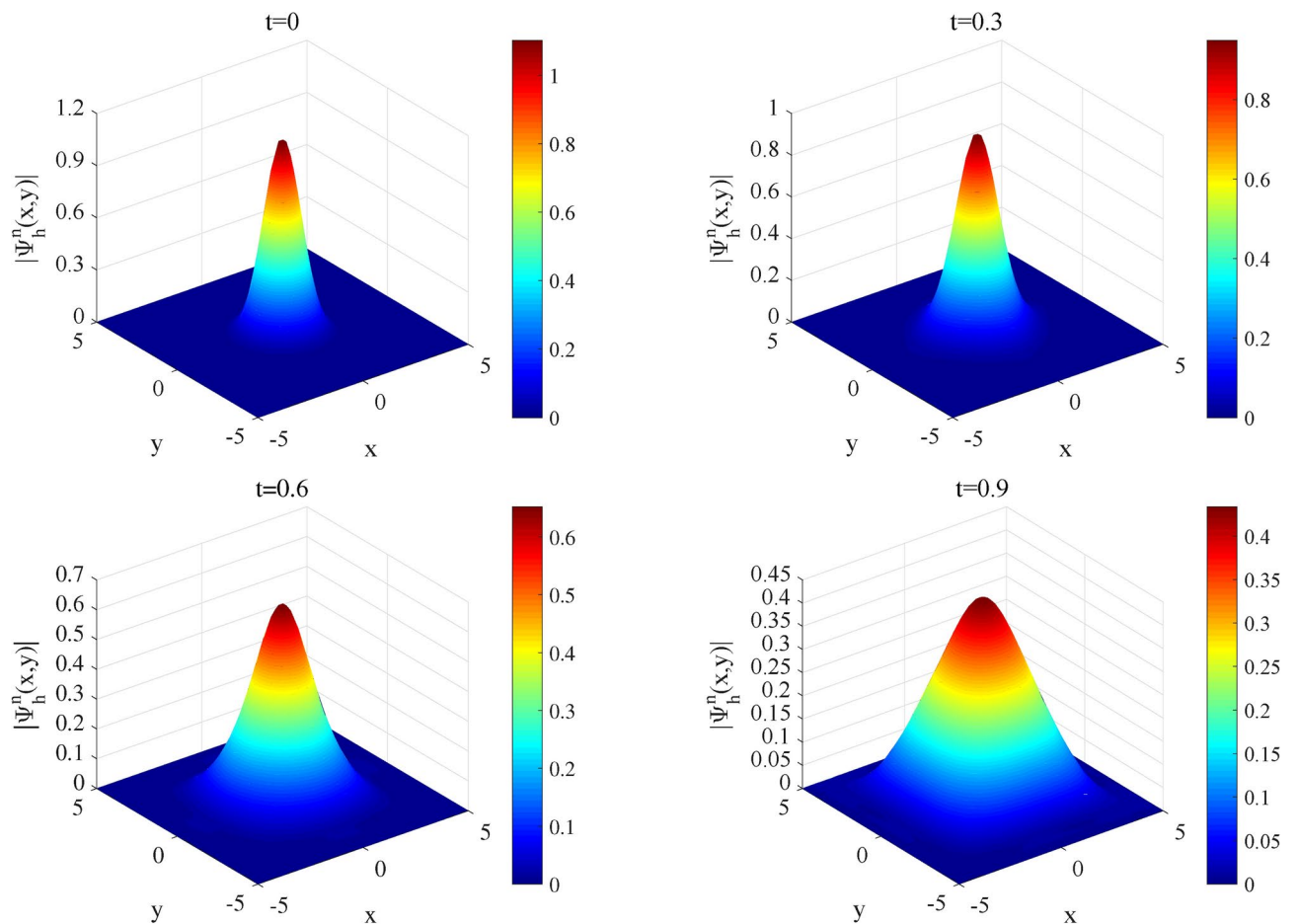


Fig. 4. The profile (left) and contour map (right) of the modulus of wave solutions for $\alpha = 0.75$.

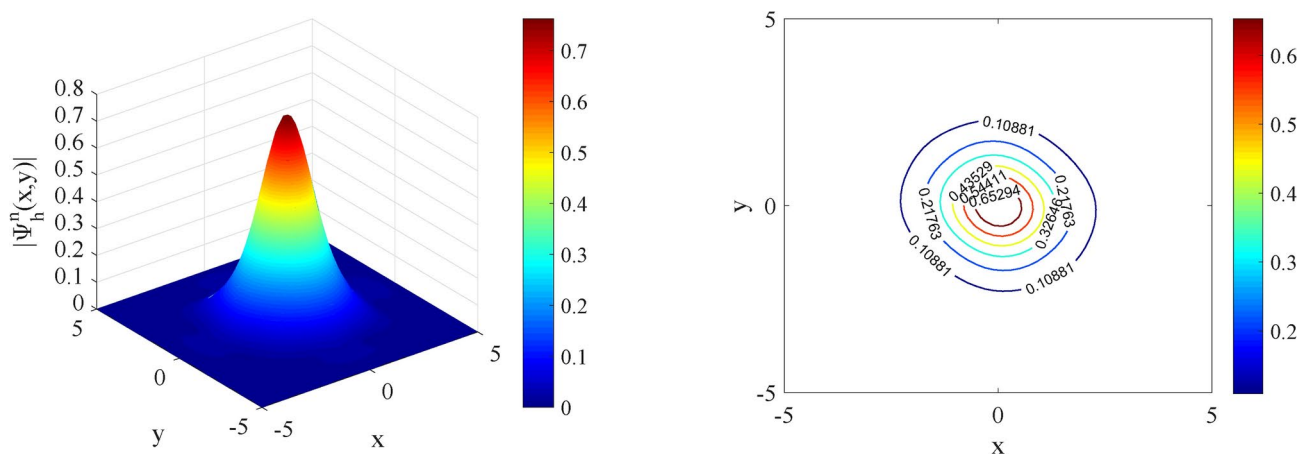


Fig. 5. The profile (left) and contour map (right) of the modulus of wave solutions for $\alpha = 0.95$.

in solving both one- and two-dimensional space-fractional NLS in practice. This split-step FEM compares favorably with the other algorithms like linearized FEM in computational efficiency and holds great potential in treating the two- and three-dimensional space-fractional NLS or CNLS on complex domains.

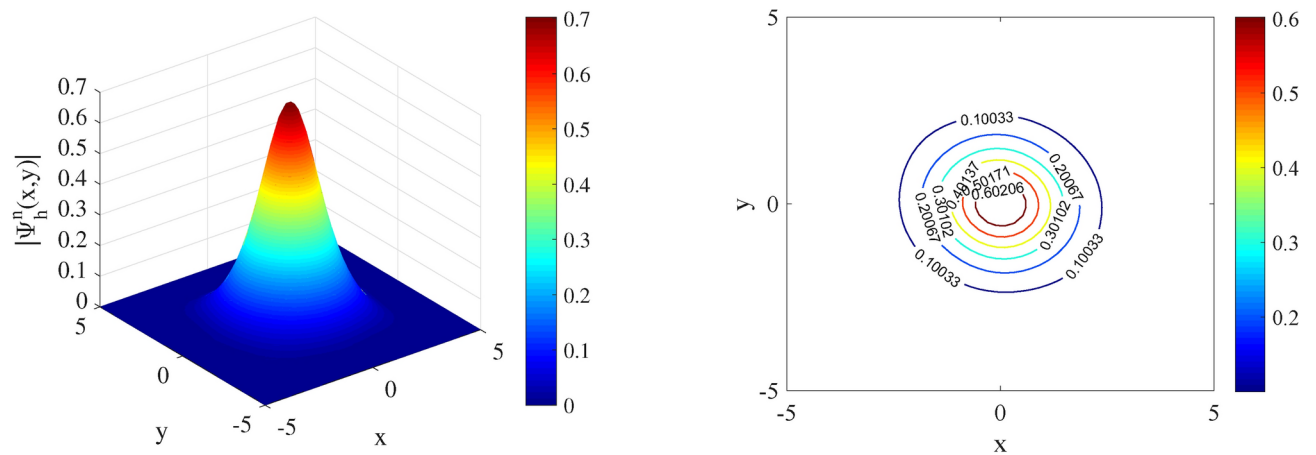


Fig. 6. The profile (left) and contour (right) of the modulus of classical wave solution.

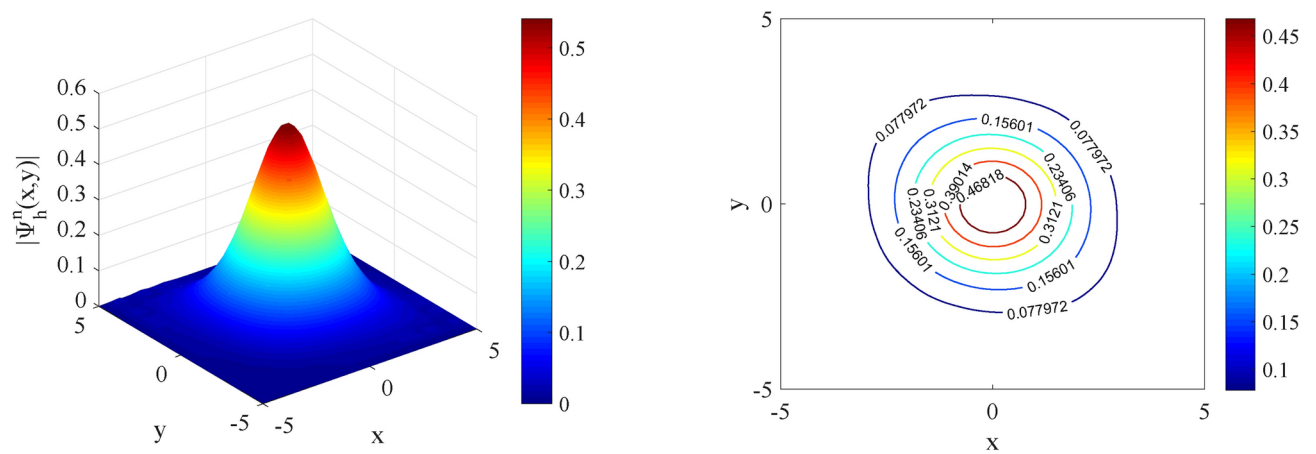


Fig. 7. The propagation of single soliton for $\alpha = 0.75, 0.85, 0.95$ and 1 with $\varepsilon = 1, \lambda = 2$.

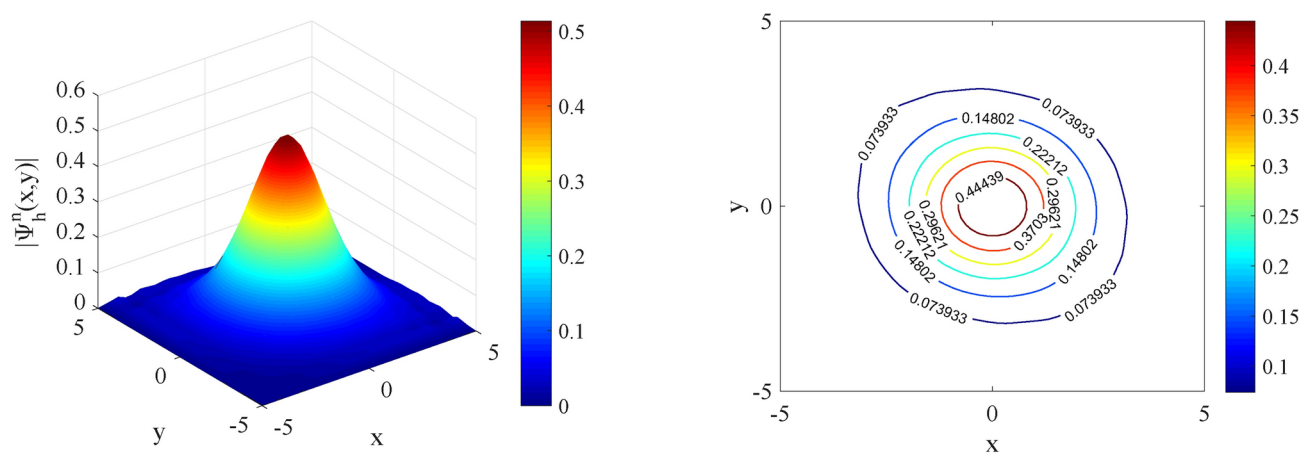


Fig. 8. The propagation of defocusing soliton for $\alpha = 0.75, 0.85, 0.95$ and 1 with $\varepsilon = 0.1, \lambda = -1$.

$\Delta t, h$	Methods	CPU time (s)
1/500, 1/3	Split-step FEM	851.16
1/500, 1/3	Crank–Nicolson FEM with Newton's iteration	1577.14
1/500, 1/3	Linearized Crank–Nicolson FEM	1094.15
1/1000, 1/2	Split-step FEM	144.70
1/1000, 1/2	Crank–Nicolson FEM with Newton's iteration	266.35
1/1000, 1/2	Linearized Crank–Nicolson FEM	152.02
1/1500, 1/4	Split-step FEM	3435.05
1/1500, 1/4	Crank–Nicolson FEM with Newton's iteration	9199.40
1/1500, 1/4	Linearized Crank–Nicolson FEM	5297.28

Table 7. The CPU times of the split-step, linearized Crank–Nicolson FEMs and the Crank–Nicolson FEM with Newton's iteration with $T = 0.5$.

Data availability

Data will be made available from the corresponding author on reasonable request.

Received: 4 June 2024; Accepted: 7 October 2024

Published online: 16 October 2024

References

1. Laskin, N. Fractional quantum mechanics. *Phys. Rev. E* **62**, 3135–3145 (2000).
2. Laskin, N. Fractional quantum mechanics and Lévy path integrals. *Phys. Lett. A* **268**, 298–305 (2000).
3. Guo, B. L. & Huo, Z. H. Global well-posedness for the fractional nonlinear Schrödinger equation. *Commun. Part. Differ. Equ.* **36**, 247–255 (2010).
4. Guo, B. L. & Li, Q. X. Existence of the global smooth solution to a fractional nonlinear Schrödinger system in atomic Bose-Einstein condensates. *J. Appl. Anal. Comput.* **5**, 793–808 (2015).
5. Hu, J. Q., Xin, J. & Lu, H. The global solution for a class of systems of fractional nonlinear Schrödinger equations with periodic boundary condition. *Comput. Math. Appl.* **62**, 1510–1521 (2011).
6. Felmer, P., Quaas, A. & Tan, J. G. Positive solutions of the nonlinear Schrödinger equation with the fractional Laplacian. *Proc. R. Soc. Edinb. A* **142**, 1237–1262 (2012).
7. Fujioka, J., Espinosa, A. & Rodríguez, R. F. Fractional optical solitons. *Phys. Lett. A* **374**, 1126–1134 (2010).
8. Guo, X. Y. & Xu, M. Y. Some physical applications of fractional Schrödinger equation. *J. Math. Phys.* **47**, 082104 (2006).
9. Luchko, Y. Fractional Schrödinger equation for a particle moving in a potential well. *J. Math. Phys.* **54**, 012111 (2013).
10. Wang, P. D. & Huang, C. M. Split-step alternating direction implicit difference scheme for the fractional Schrödinger equation in two dimensions. *Comput. Math. Appl.* **71**, 1114–1128 (2016).
11. Aruna, K. & Kanth, A. S. V. R. Approximate solutions of non-linear fractional Schrödinger equation via differential transform method and modified differential transform method. *Natl. Acad. Sci. Lett.* **36**, 201–213 (2013).
12. Herzallah, M. A. E. & Gepreel, K. A. Approximate solution to the time-space fractional cubic nonlinear Schrödinger equation. *Appl. Math. Model.* **36**, 5678–5685 (2012).
13. Wang, D. L., Xiao, A. G. & Yang, W. Crank-Nicolson difference scheme for the coupled nonlinear Schrödinger equations with the Riesz space fractional derivative. *J. Comput. Phys.* **242**, 670–681 (2013).
14. Ran, M. H. & Zhang, C. J. A conservative difference scheme for solving the strongly coupled nonlinear fractional Schrödinger equations. *Commun. Nonlinear Sci.* **41**, 64–83 (2016).
15. Wang, P. D. & Huang, C. M. An energy conservative difference scheme for the nonlinear fractional Schrödinger equations. *J. Comput. Phys.* **293**, 238–251 (2015).
16. Li, M., Gu, X. M., Huang, C. M., Fei, M. F. & Zhang, G. Y. A fast linearized conservative finite element method for the strongly coupled nonlinear fractional Schrödinger equations. *J. Comput. Phys.* **358**, 256–282 (2018).
17. Li, M. & Zhao, Y. L. A fast energy conserving finite element method for the nonlinear fractional Schrödinger equation with wave operator. *Appl. Math. Comput.* **338**, 758–773 (2018).
18. Zhu, X. G., Nie, Y. F., Yuan, Z. B., Wang, J. G. & Yang, Z. Z. A Galerkin FEM for Riesz space-fractional CNLS. *Adv. Differ. Equ.* **2019**, 329 (2019).
19. Zhu, X. G., Yuan, Z. B., Wang, J. G., Nie, Y. F. & Yang, Z. Z. Finite element method for time-space-fractional Schrödinger equation. *Electron. J. Differ. Equ.* **166**, 1–18 (2017).
20. Zhao, X., Sun, Z. Z. & Hao, Z. P. A fourth-order compact ADI scheme for two-dimensional nonlinear space fractional Schrödinger equation. *SIAM J. Sci. Comput.* **36**, A2865–A2886 (2014).
21. Fan, W. P. & Qi, H. T. An efficient finite element method for the two-dimensional nonlinear time-space fractional Schrödinger equation on an irregular convex domain. *Appl. Math. Lett.* **86**, 103–110 (2018).
22. Zhang, R. P., Zhang, Y. T., Wang, Z., Chen, B. & Zhang, Y. A conservative numerical method for the fractional nonlinear Schrödinger equation in two dimensions. *Sci. China Math.* **62**, 1997–2014 (2019).
23. Wang, Y., Mei, L. Q., Li, Q. & Bu, L. L. Split-step spectral Galerkin method for the two-dimensional nonlinear space-fractional Schrödinger equation. *Appl. Numer. Math.* **136**, 257–278 (2019).
24. Wang, W. S., Huang, Y. & Tang, J. Lie-Trotter operator splitting spectral method for linear semiclassical fractional Schrödinger equation. *Comput. Math. Appl.* **113**, 117–129 (2022).
25. Abdolabadi, F., Zakeri, A. & Amiraslani, A. A split-step Fourier pseudo-spectral method for solving the space fractional coupled nonlinear Schrödinger equations. *Commun. Nonlinear Sci.* **120**, 107150 (2023).
26. Frutos, J. D. & Sanz-Serna, J. M. Split-step spectral schemes for nonlinear Dirac systems. *J. Comput. Phys.* **83**, 407–423 (1989).
27. Li, L. F. A split-step finite-element method for incompressible Navier-Stokes equations with high-order accuracy up-to the boundary. *J. Comput. Phys.* **408**, 109274 (2020).
28. Ervin, V. J. & Roop, J. P. Variational formulation for the stationary fractional advection dispersion equation. *Numer. Methods Part D. E.* **22**, 558–576 (2006).
29. Roop, J. P. Computational aspects of FEM approximation of fractional advection dispersion equations on bounded domains in $\mathbb{R}^2$. *J. Comput. Appl. Math.* **193**, 243–268 (2006).

30. Zhu, X. G., Nie, Y. F., Wang, J. G. & Yuan, Z. B. A numerical approach for the Riesz space-fractional Fisher' equation in two-dimensions. *Int. J. Comput. Math.* **94**, 296–315 (2017).
31. Bu, W. P., Tang, Y. F. & Yang, J. Y. Galerkin finite element method for two-dimensional Riesz space fractional diffusion equations. *J. Comput. Phys.* **276**, 26–38 (2014).
32. Gauckler, L. Convergence of a split-step Hermite method for the Gross-Pitaevskii equation. *IMA J. Numer. Anal.* **31**, 396–415 (2011).
33. Zhou, Y. *Application of Discrete Functional Analysis to the Finite Difference Methods* (International Academic Publishers, 1990).
34. Zhai, S. Y., Wang, D. L., Weng, Z. F. & Zhao, X. Error analysis and numerical simulations of Strang splitting method for space fractional nonlinear Schrödinger equation. *J. Sci. Comput.* **81**, 965–989 (2019).
35. Lubich, C. On splitting methods for Schrödinger-Poisson and cubic nonlinear Schrödinger equations. *Math. Comput.* **77**, 2141–2153 (2008).
36. Duo, S. W. & Zhang, Y. Z. Mass-conservative Fourier spectral methods for solving the fractional nonlinear Schrödinger equation. *Comput. Math. Appl.* **71**, 2257–2271 (2016).

Acknowledgements

This research was supported by the Science and Technology Planning Projects of Shaoyang (No. 2023ZD0074), the Scientific Research Funds of Hunan Provincial Education Department (Nos. 22B0756, 22C0455 and 21C0607) and the Natural Science Foundation of Hunan Province of China (No. 2022JJ30548).

Author contributions

X.Z.: methodology, formal analysis, investigation, data curation, writing—original draft; H.W.: supervision, writing—review and editing; Y.Z.: writing—review and editing. All authors read and approved the final manuscript.

Declarations

Competing interests

The authors declare no competing interests.

Additional information

Correspondence and requests for materials should be addressed to X.Z.

Reprints and permissions information is available at www.nature.com/reprints.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

© The Author(s) 2024